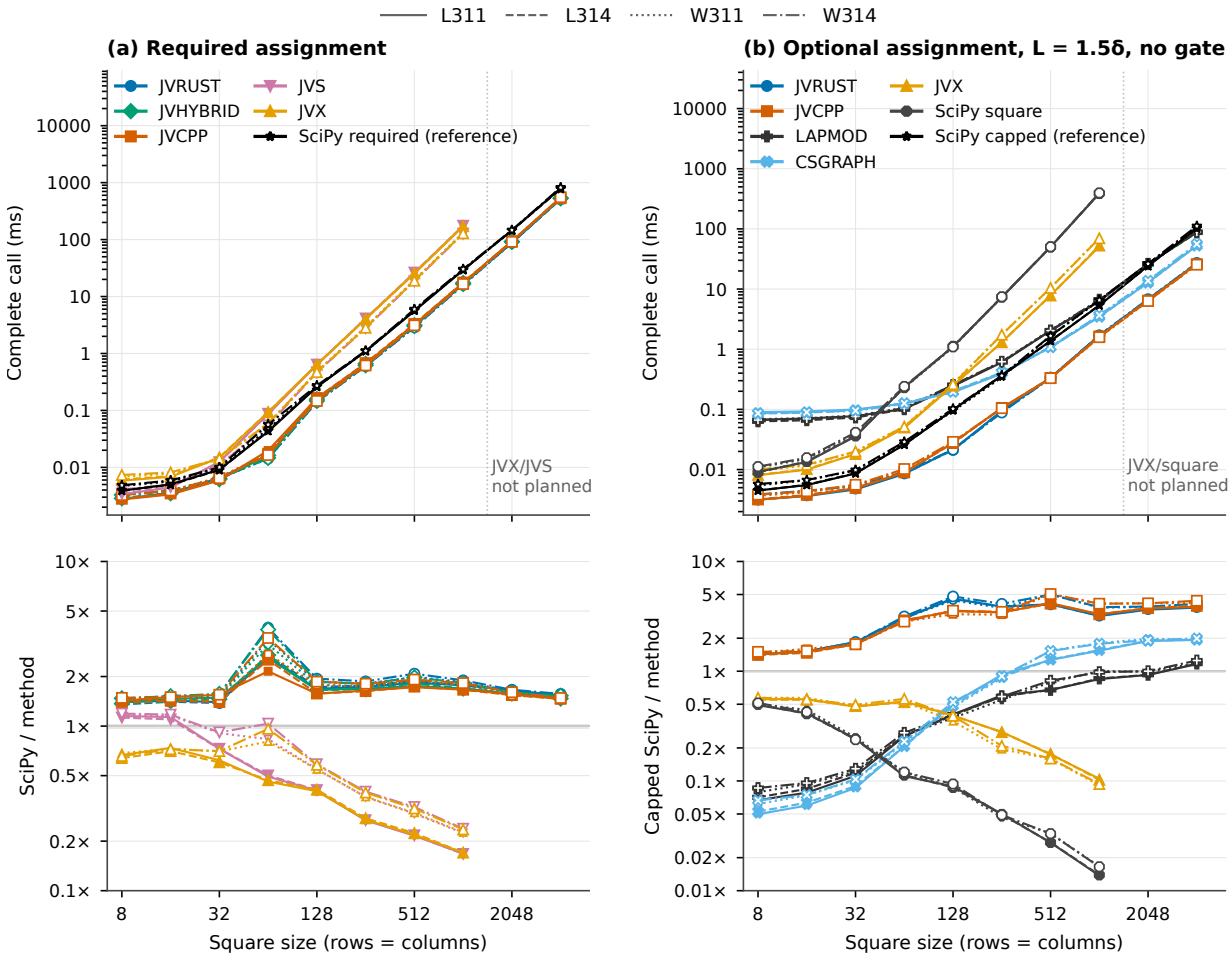
Native Rectangular Assignment: Saved-Result Figure Atlas

This supplement shows ordered workload families and named controlled comparisons from the four saved light sessions. Each curve or marker represents one environment. The four line styles identify L311 (solid), L314 (dashed), W311 (dotted) and W314 (dash-dot). Markers are filled for Linux and open for Windows; smaller markers identify Python 3.11 and larger markers Python 3.14. Each figure keys the environments explicitly. Coincident values can still overlap. Colors identify methods consistently with the main paper. Gray bands mark ratios in $[0.97, 1.03]$; classifications use unrounded values. Missing observations mean "not planned", never zero. Policy ratios are intervention/reference (above one is slower); other ratio axes state their direction. Exact case values, attempt identities and saved validation outcomes remain in the indexed CSV exports; environments are never averaged.

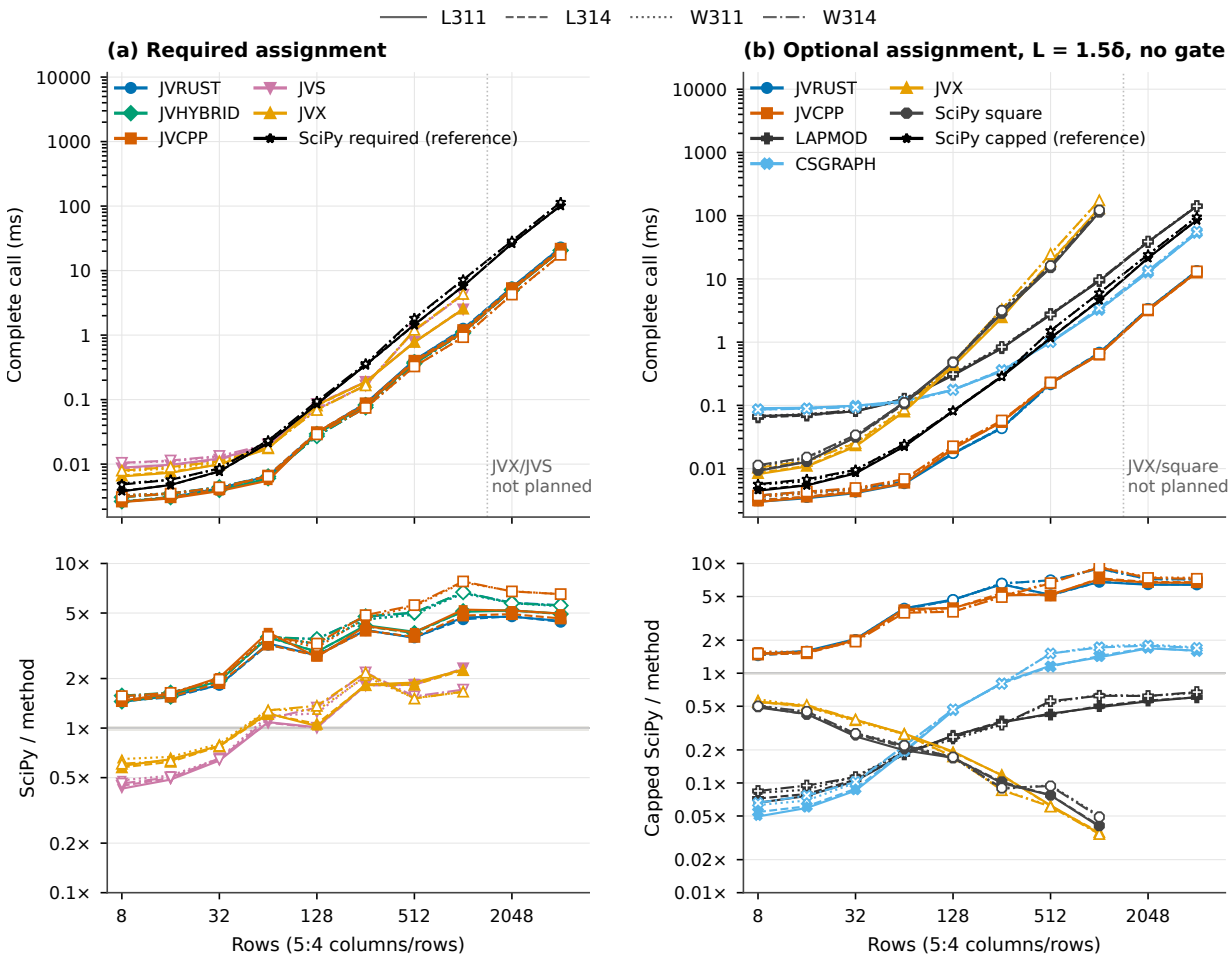
Atlas 1. Geometry: q=0.6, sd=1δ, square

Ten saved sizes per objective. Runtime above, matching SciPy/method ratios below. Each curve is one method/environment. Only planned methods are connected; the light cutoff is marked.



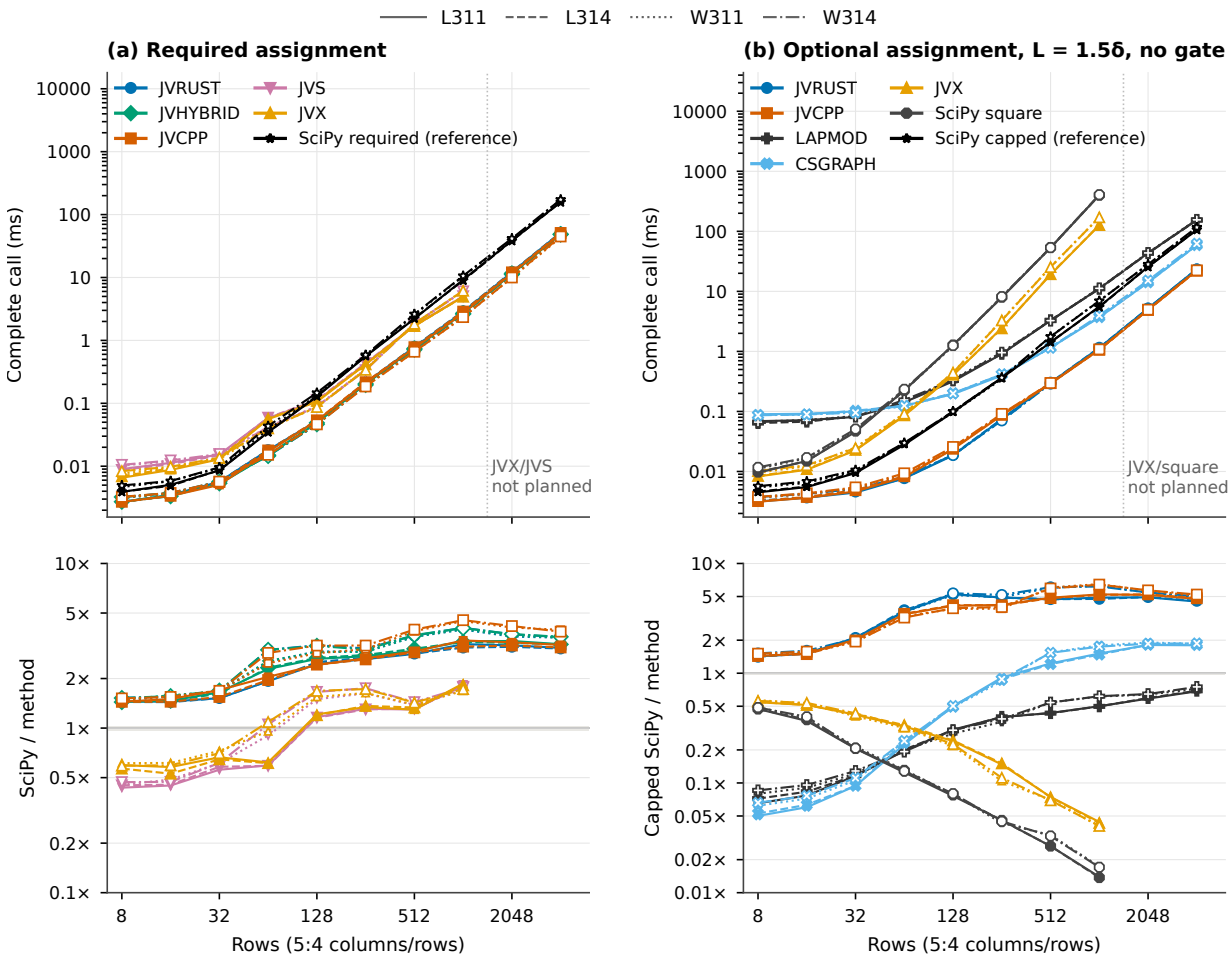
Atlas 2. Geometry: q=0.9, sd=0.25δ, 5:4

Ten saved sizes per objective. Runtime above, matching SciPy/method ratios below. Each curve is one method/environment. Only planned methods are connected; the light cutoff is marked.



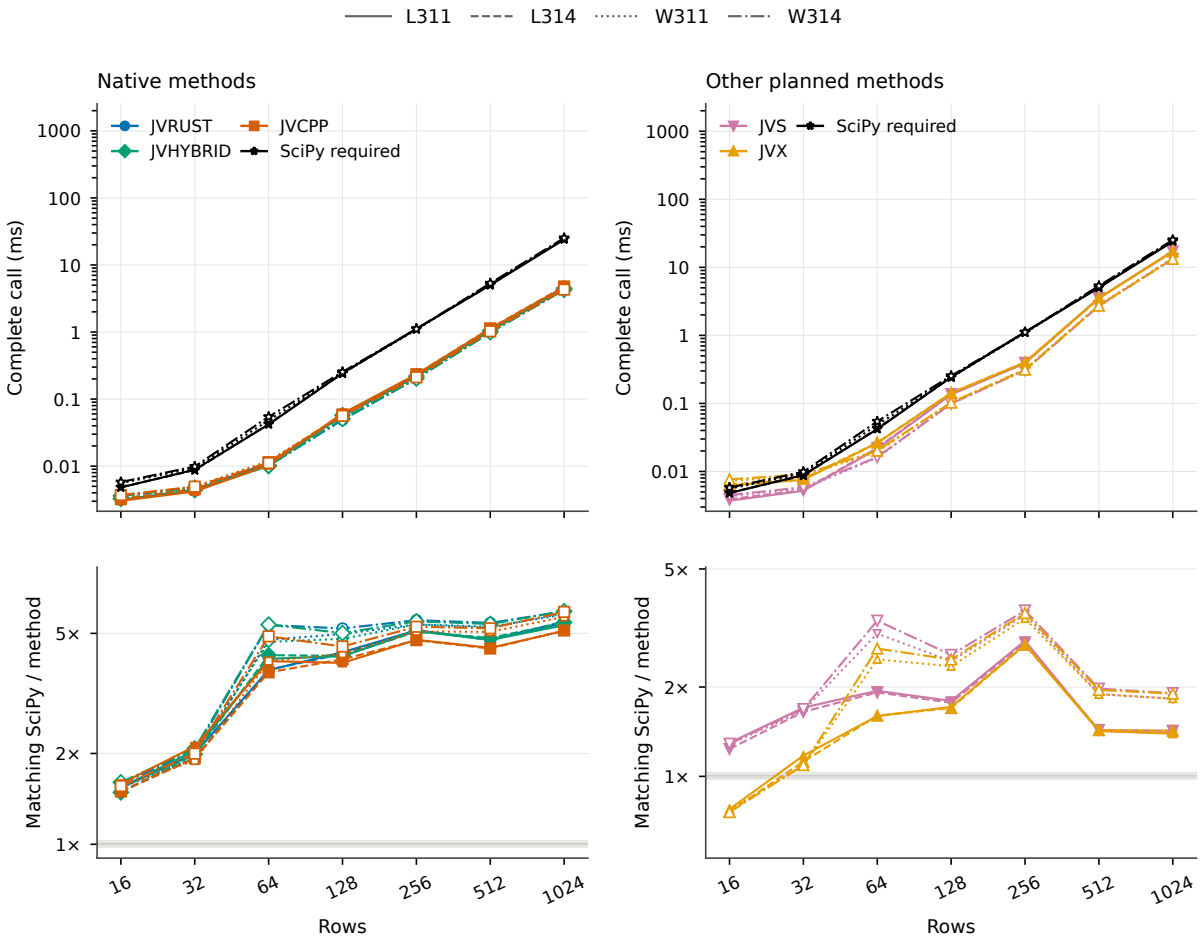
Atlas 3. Geometry: q=0.6, sd=1δ, 5:4

Ten saved sizes per objective. Runtime above, matching SciPy/method ratios below. Each curve is one method/environment. Only planned methods are connected; the light cutoff is marked.



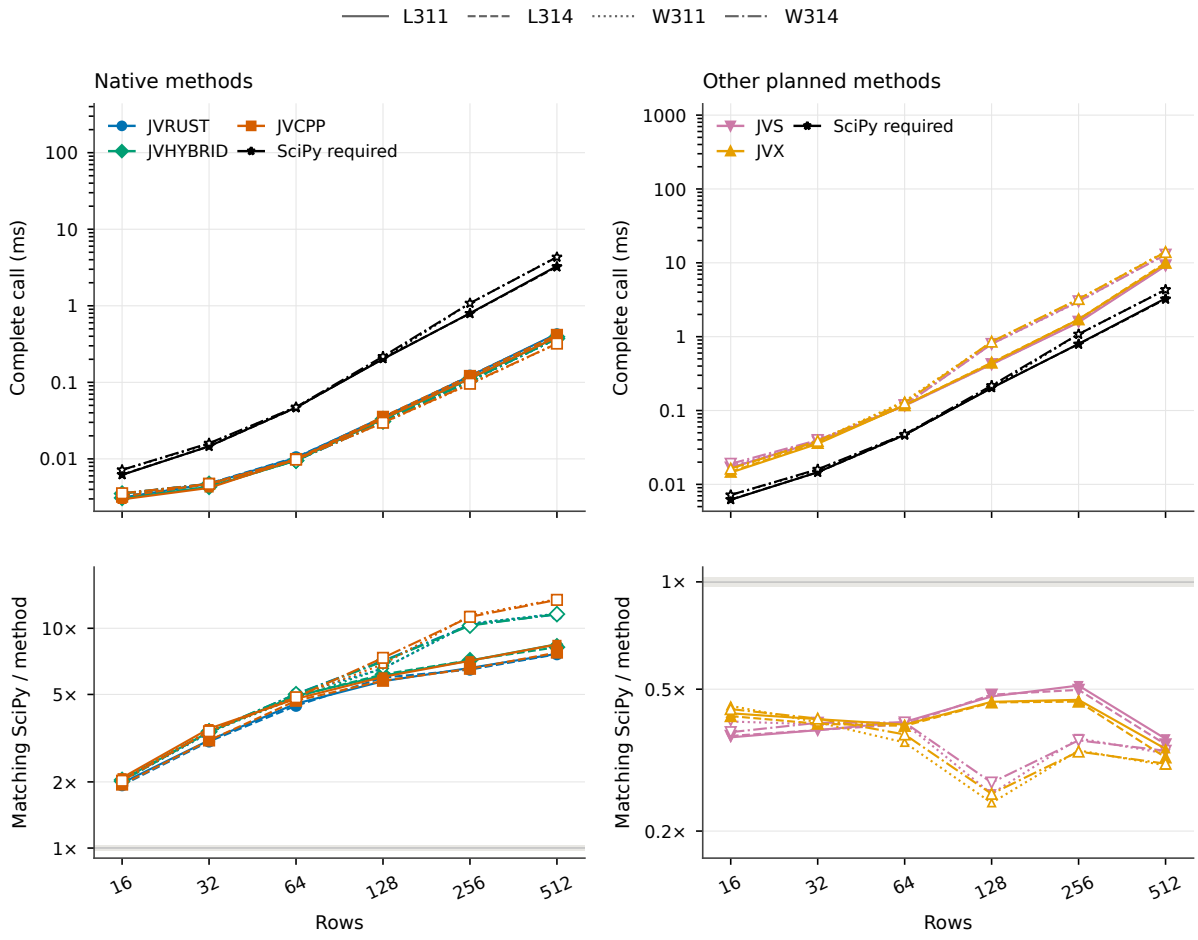
Atlas 4. Required square

7 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



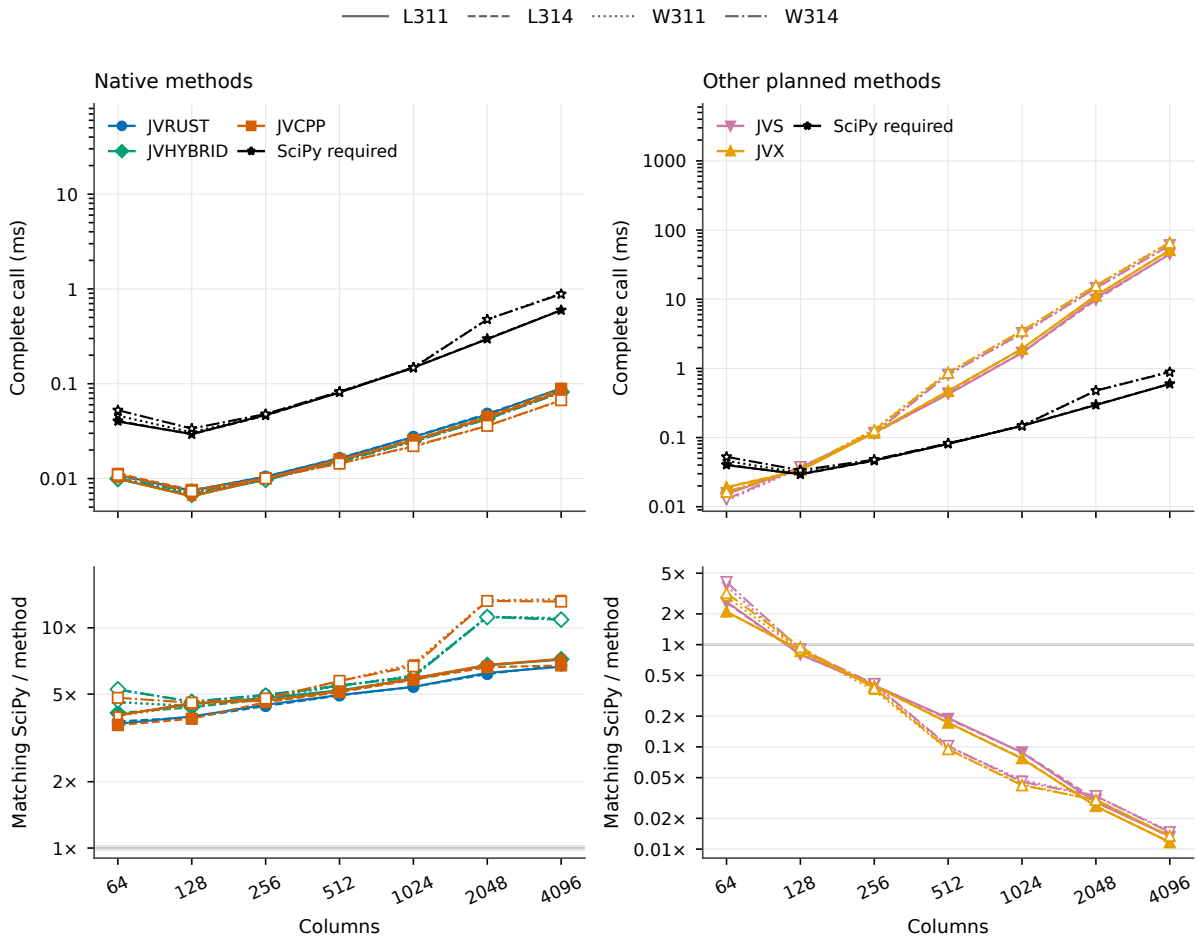
Atlas 5. Required wide

6 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



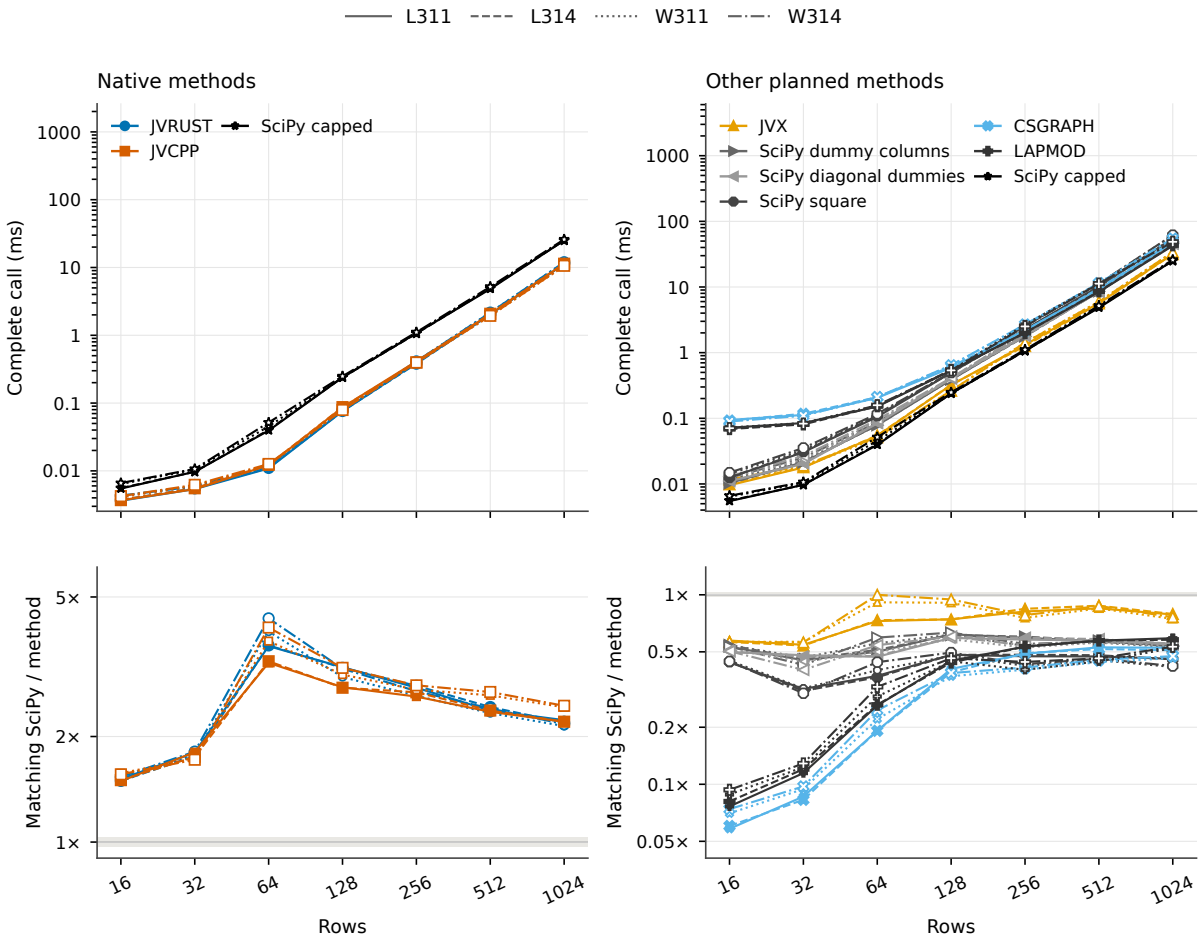
Atlas 6. Required fixed

7 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



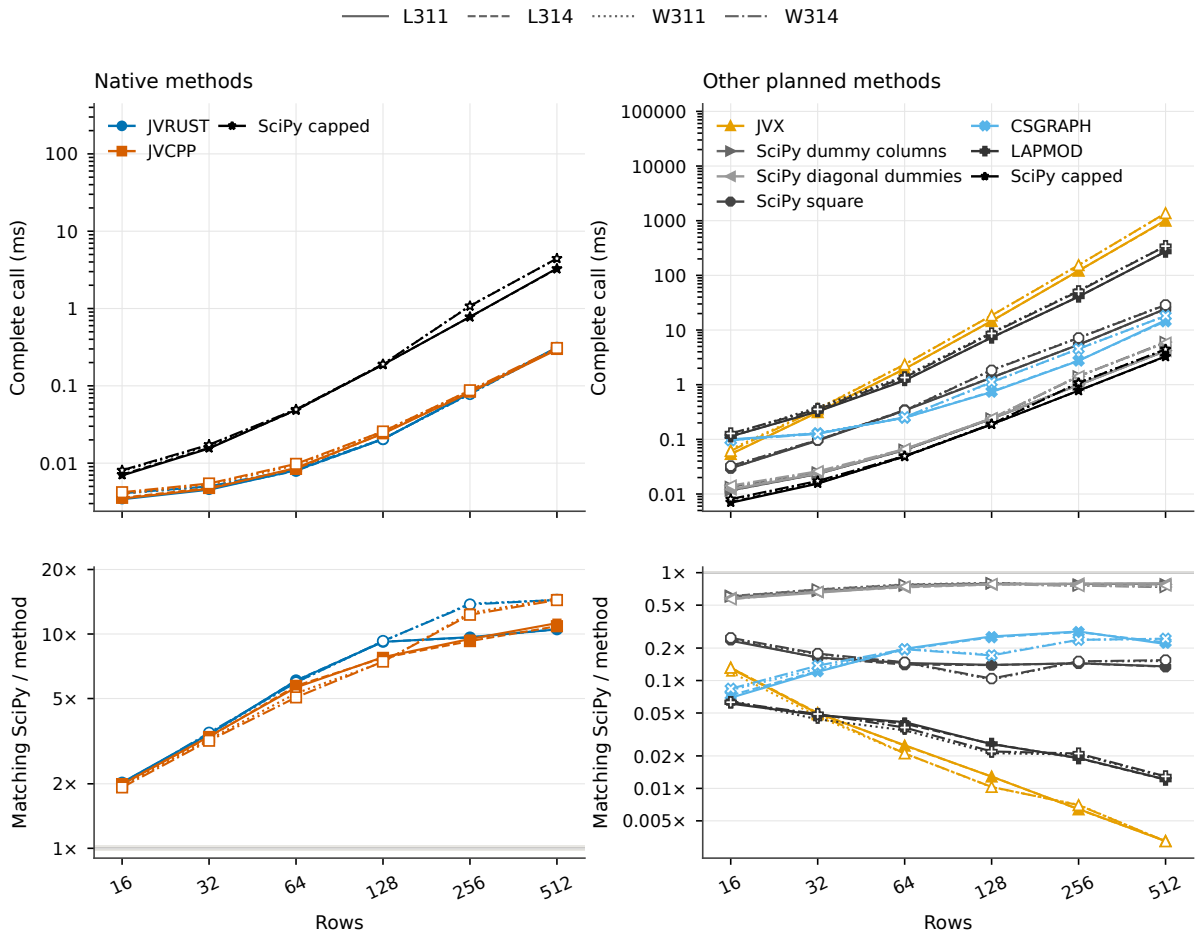
Atlas 7. Optional square

7 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



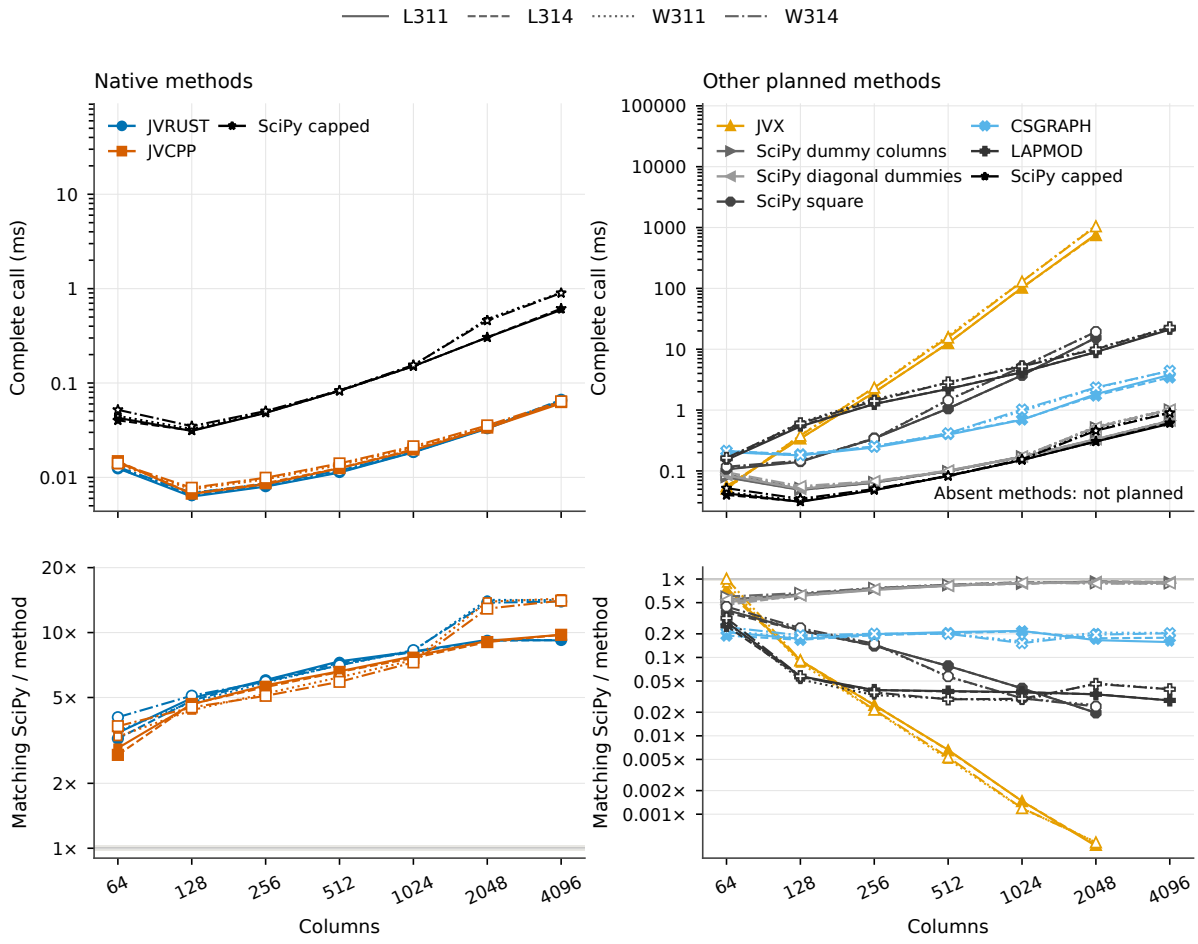
Atlas 8. Optional wide

6 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



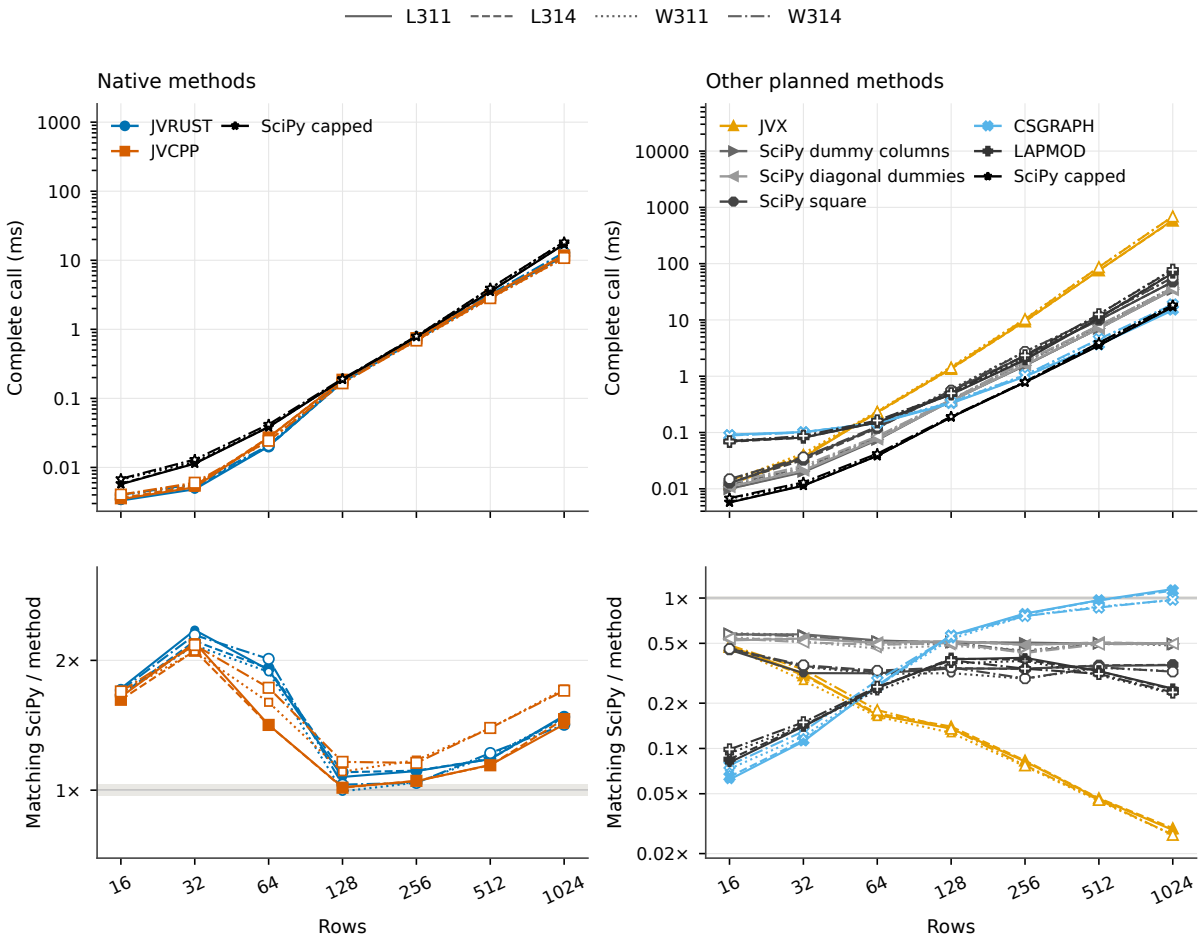
Atlas 9. Optional fixed

7 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



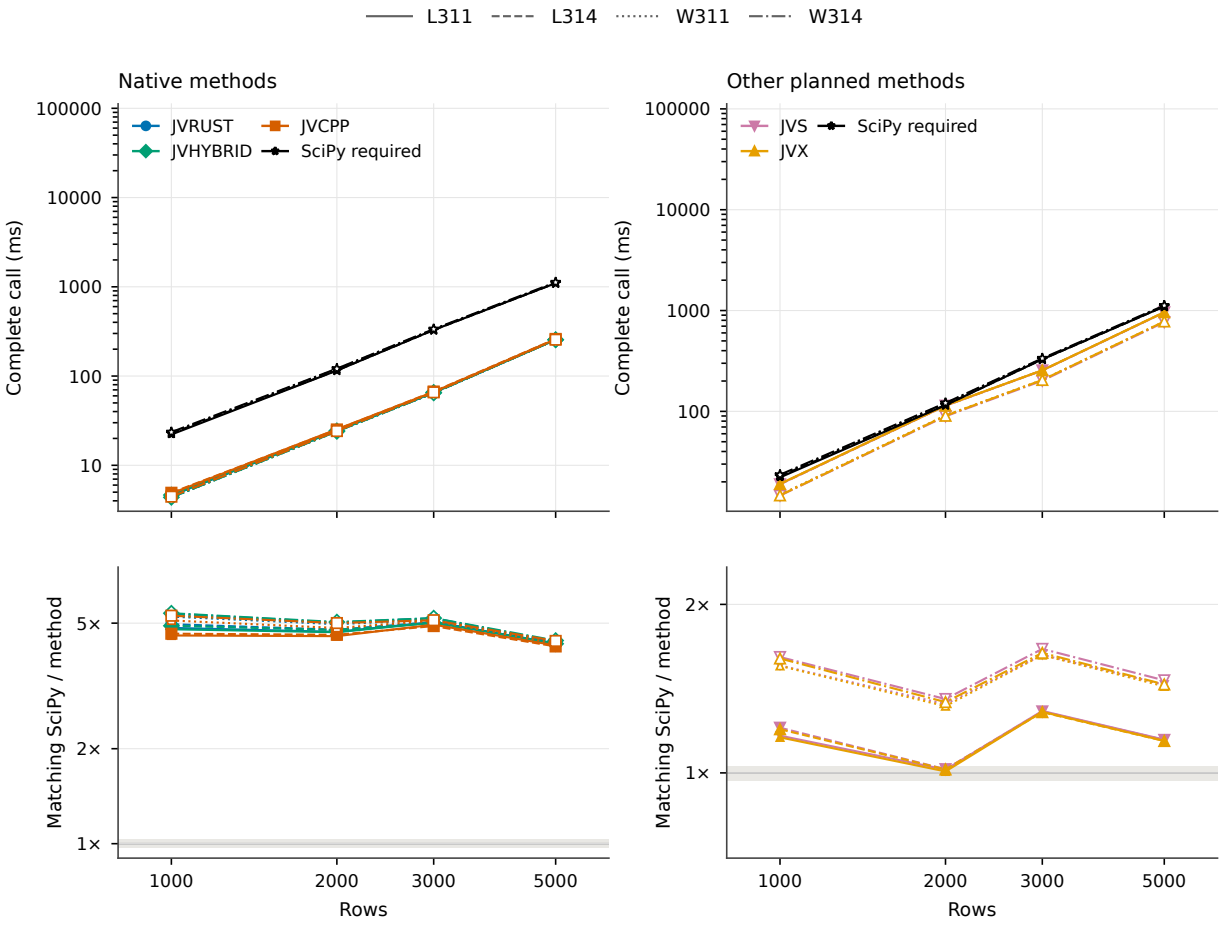
Atlas 10. Optional blocks

7 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



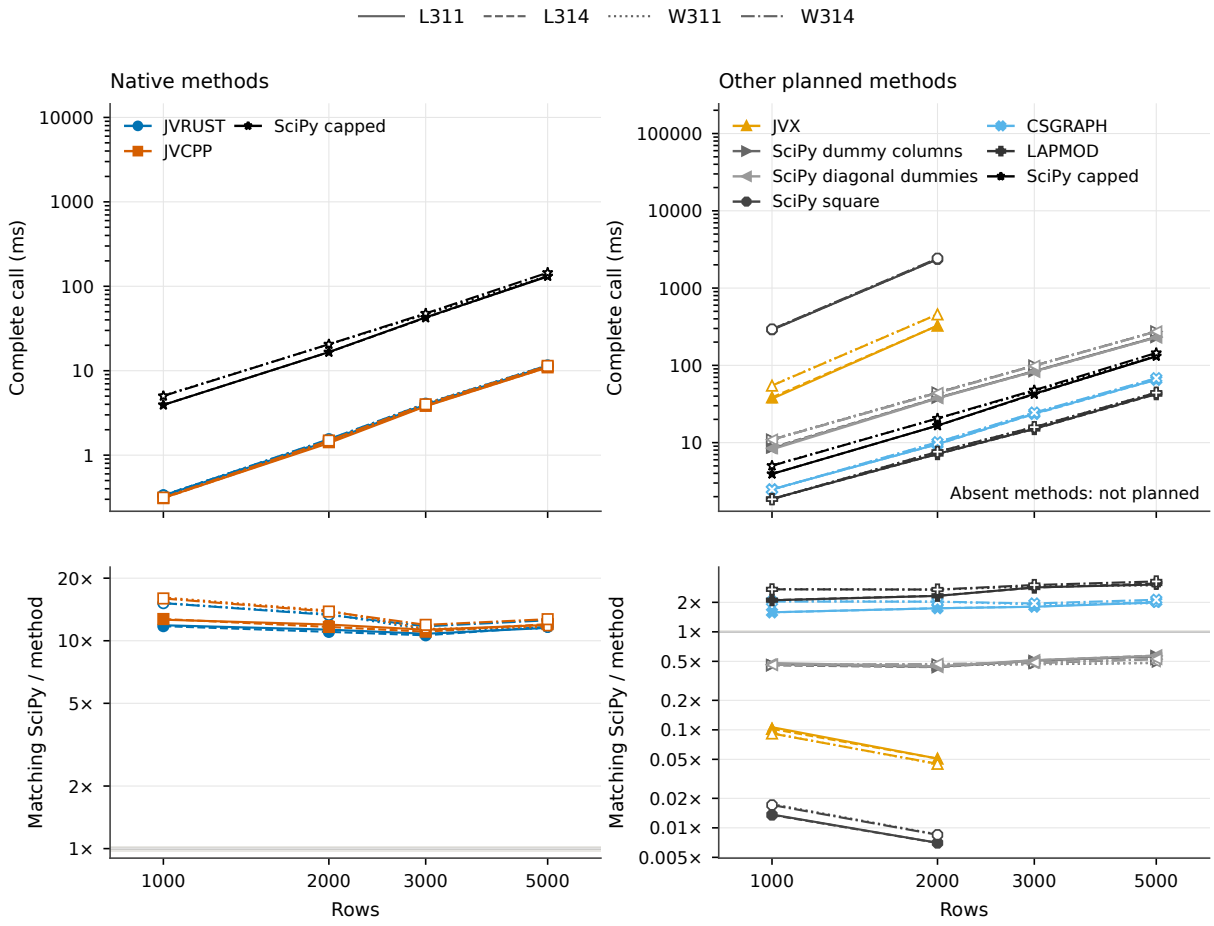
Atlas 11. Large square required

4 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



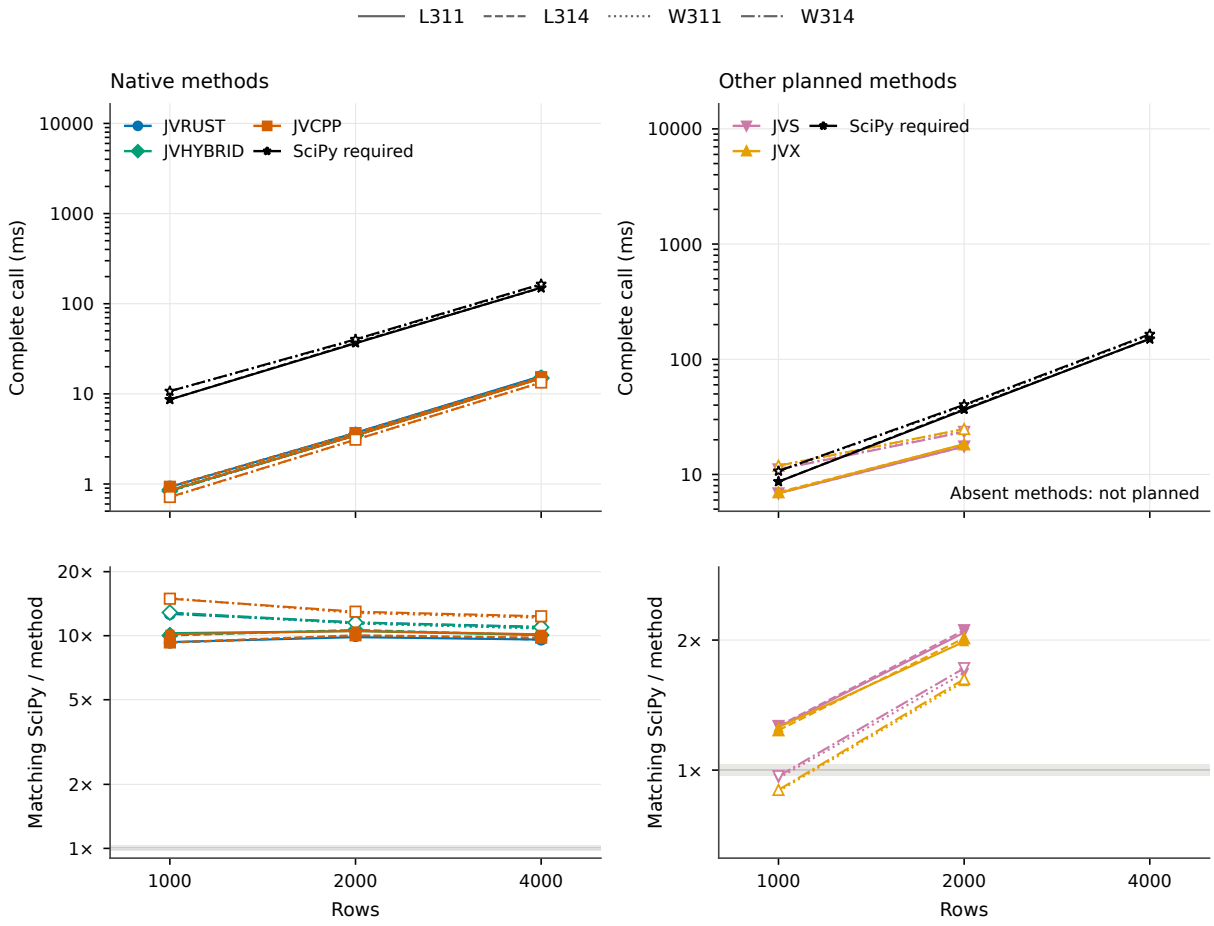
Atlas 12. Large square optional

4 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



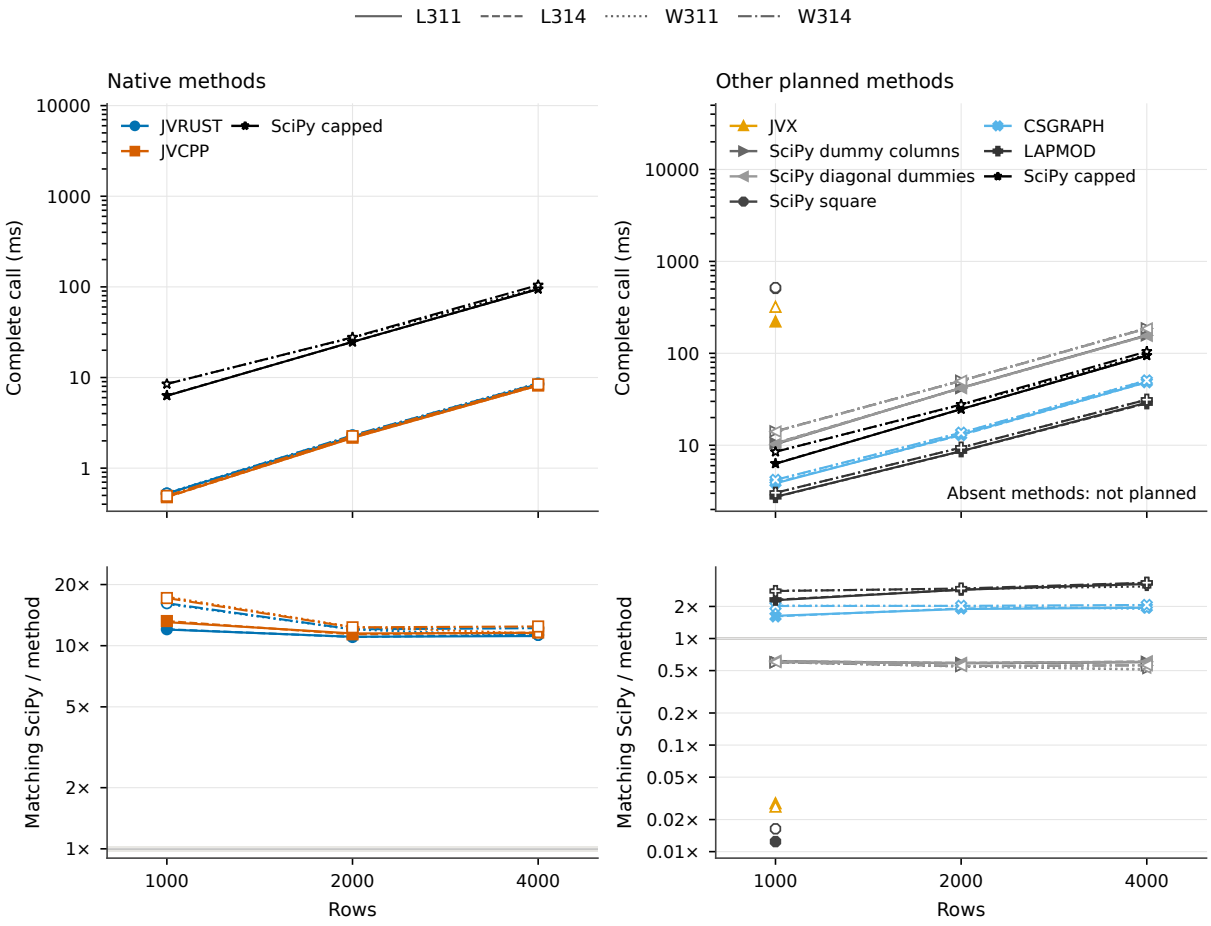
Atlas 13. Large rectangle required wide

3 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



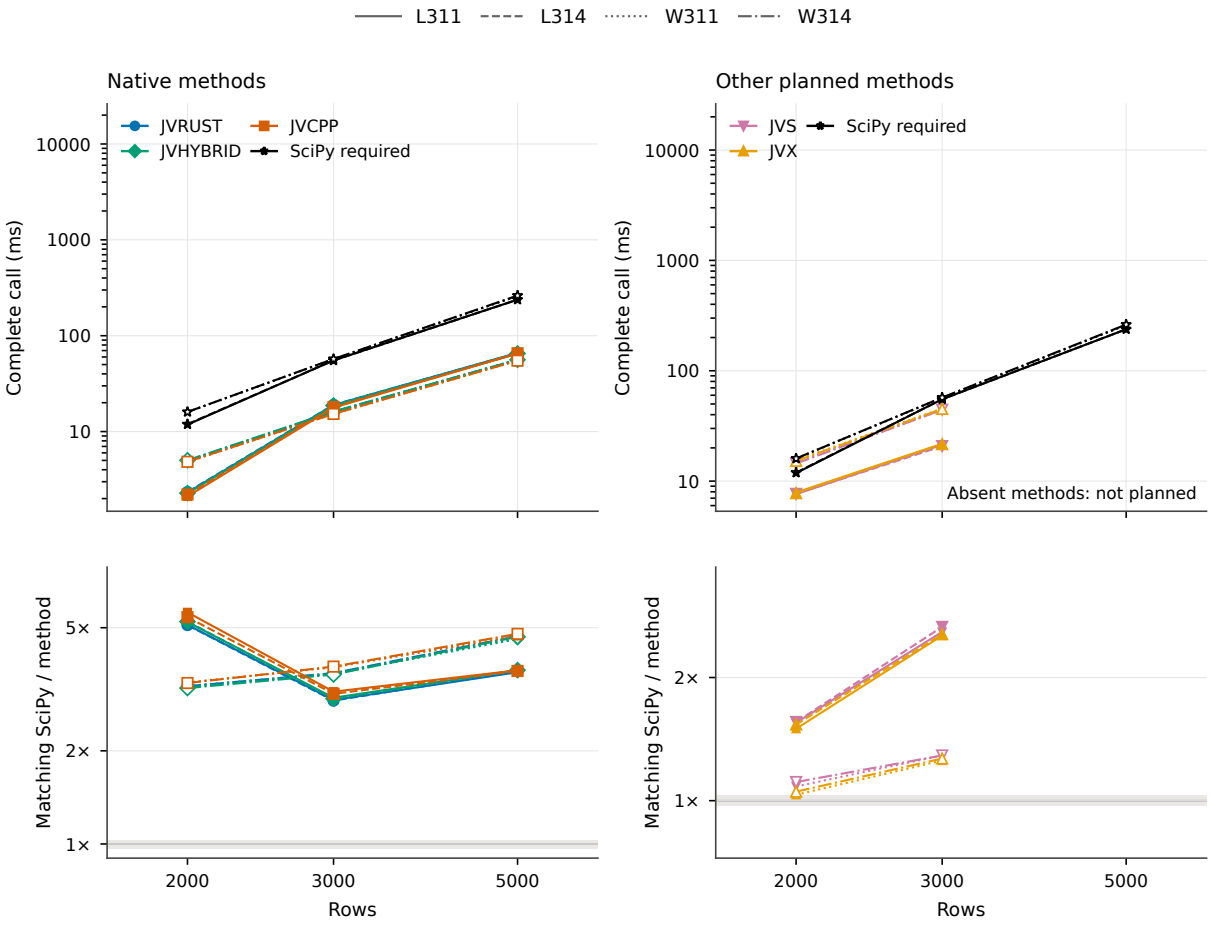
Atlas 14. Large rectangle optional wide

3 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



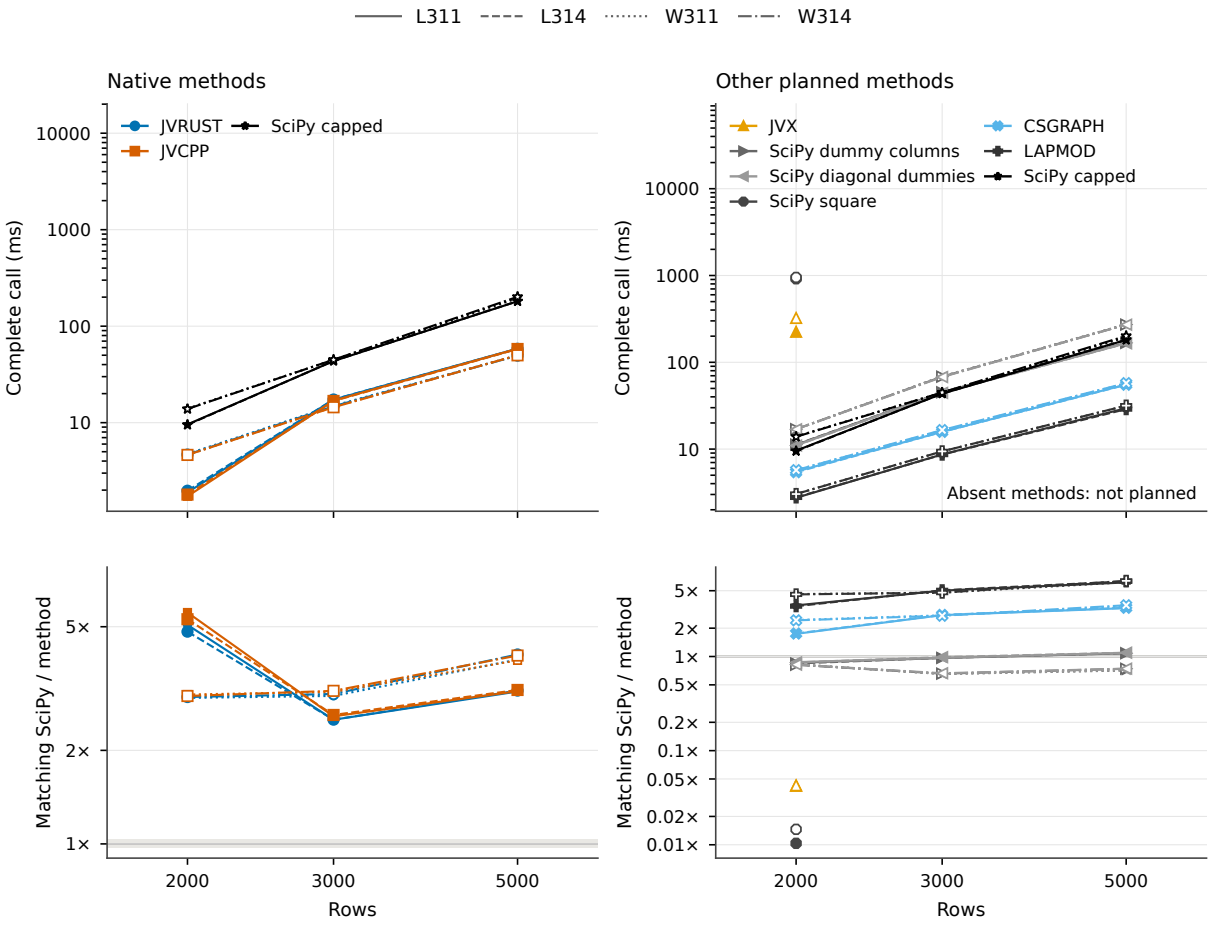
Atlas 15. Large rectangle required tall

3 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



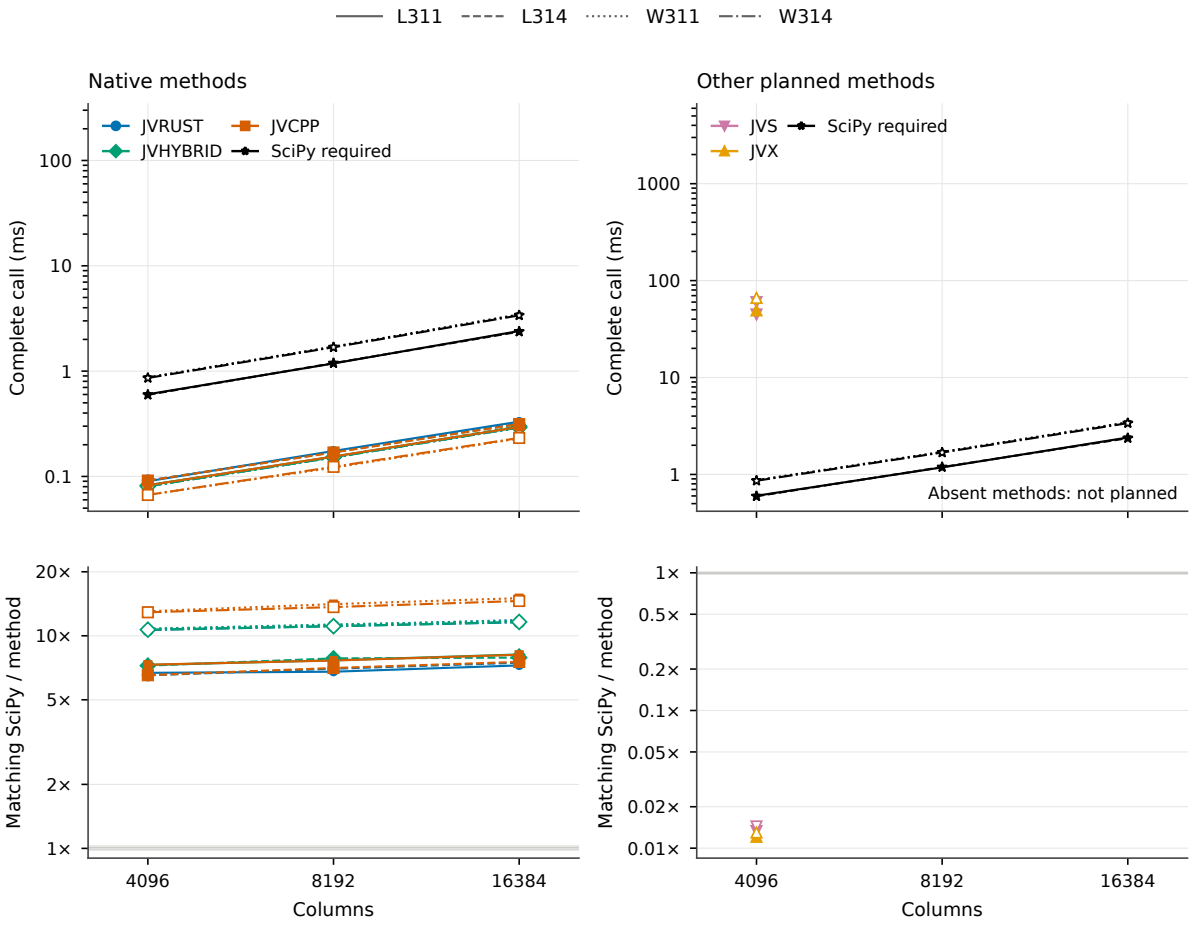
Atlas 16. Large rectangle optional tall

3 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



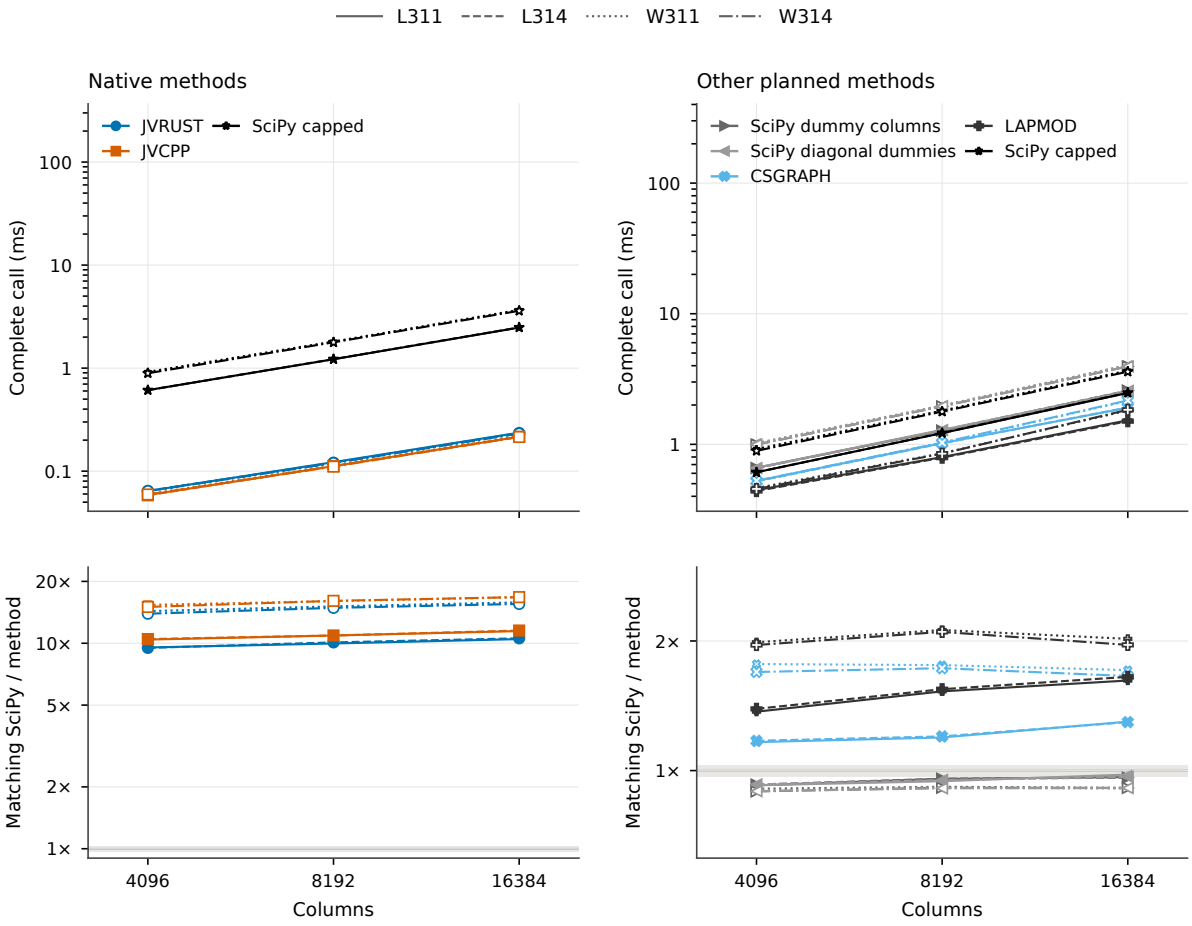
Atlas 17. Extreme rectangle required wide

3 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



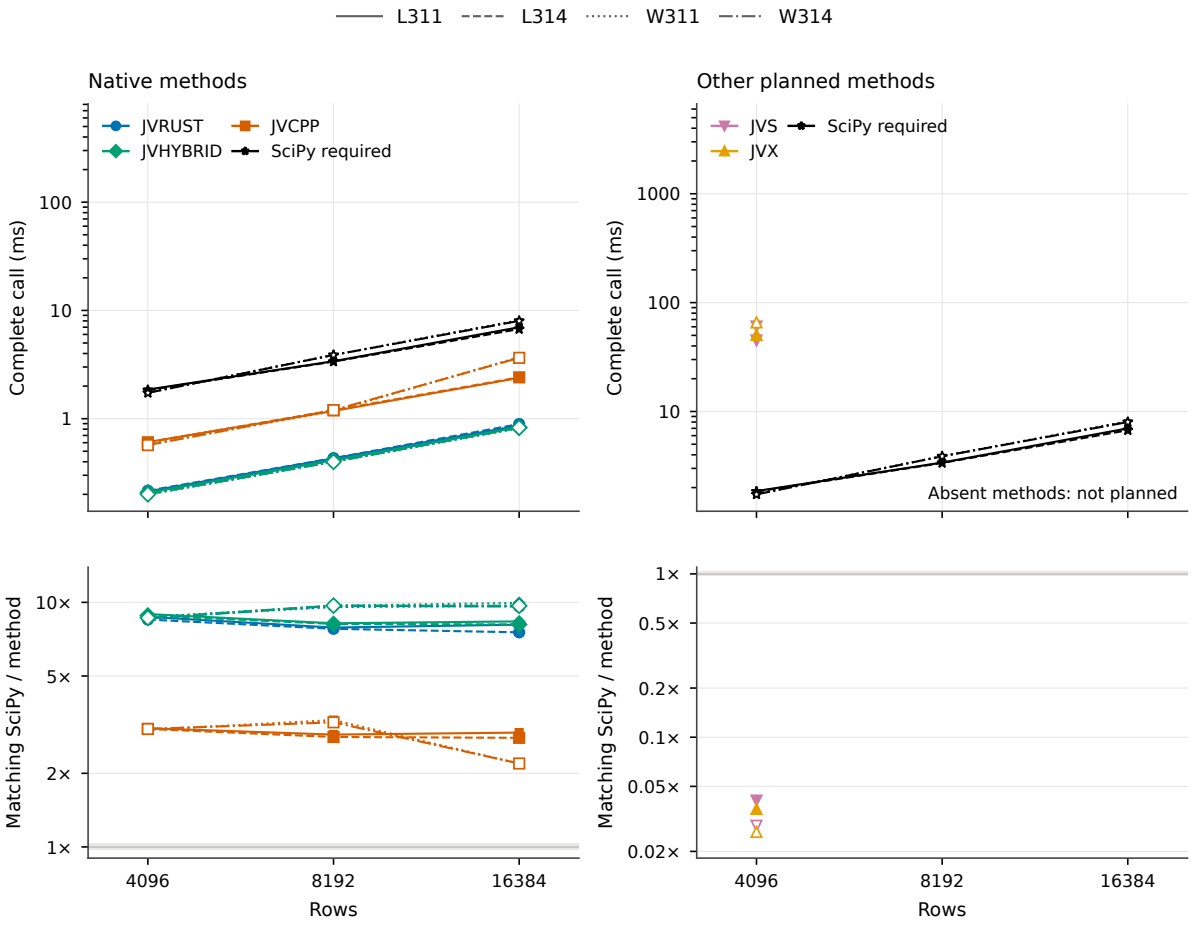
Atlas 18. Extreme rectangle optional wide

3 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



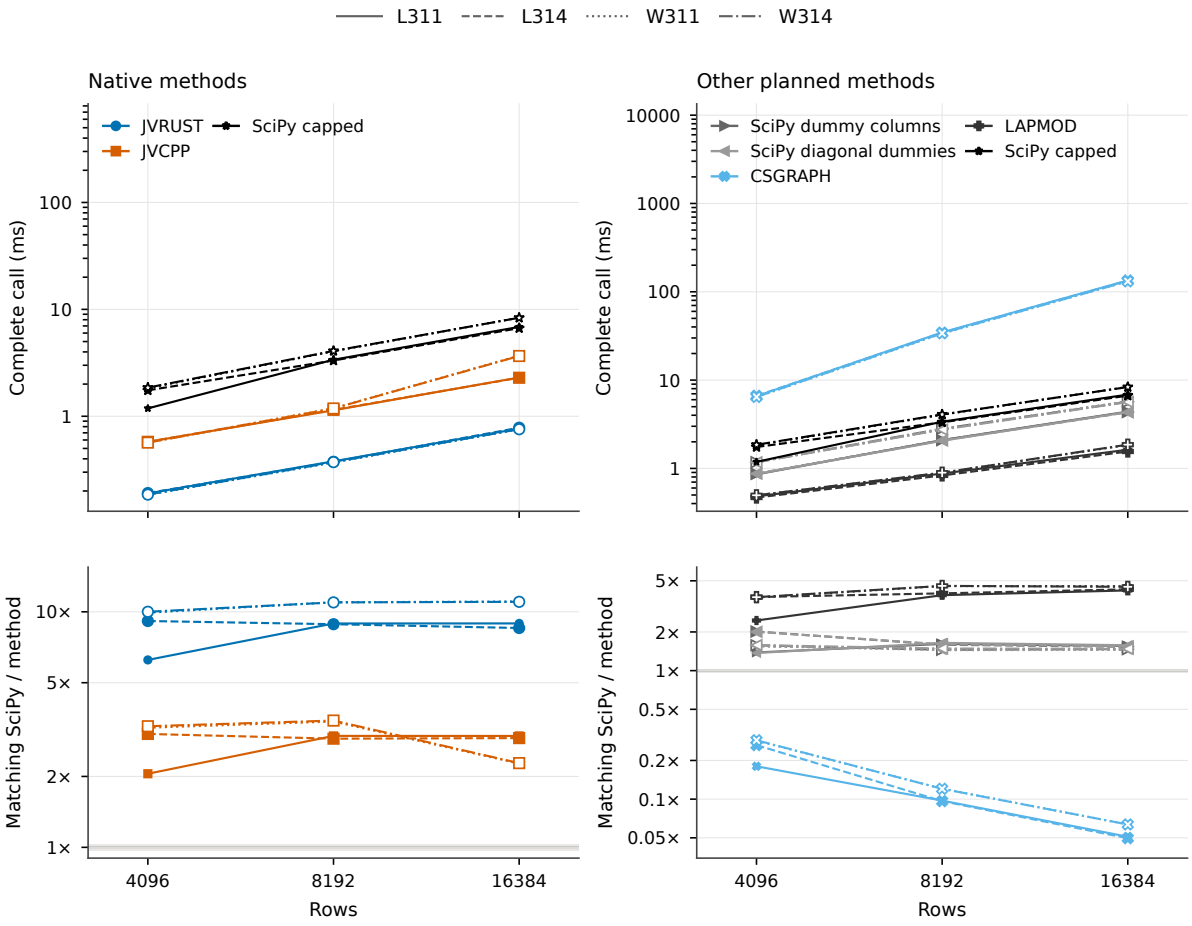
Atlas 19. Extreme rectangle required tall

3 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



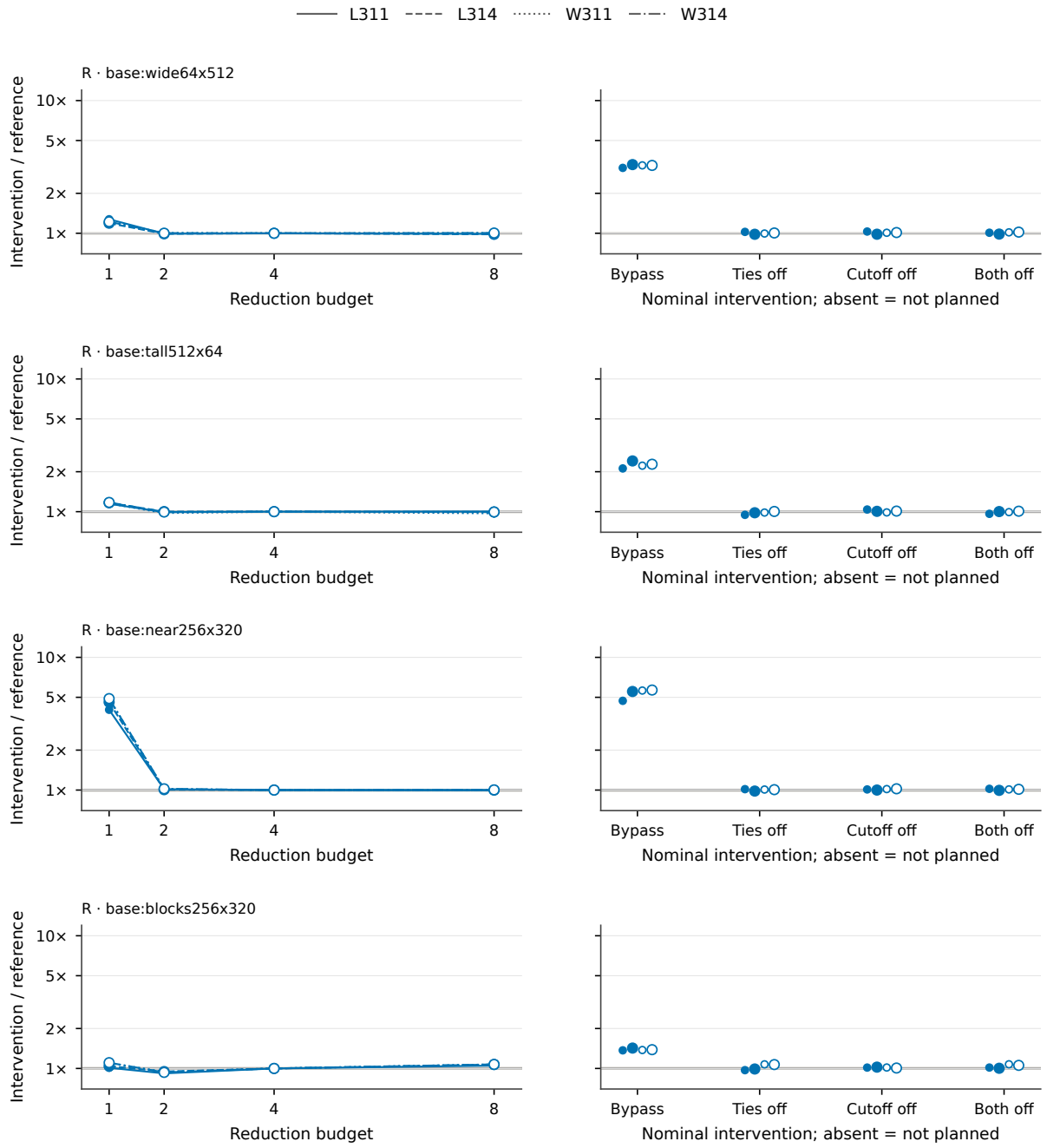
Atlas 20. Extreme rectangle optional tall

3 saved sizes. Top: complete-call milliseconds. Bottom: matching SciPy/method ratio. Each curve is one method and one environment, distinguished by four keyed line styles. Only planned observations are drawn. Exact values are in A1_scaling.csv.



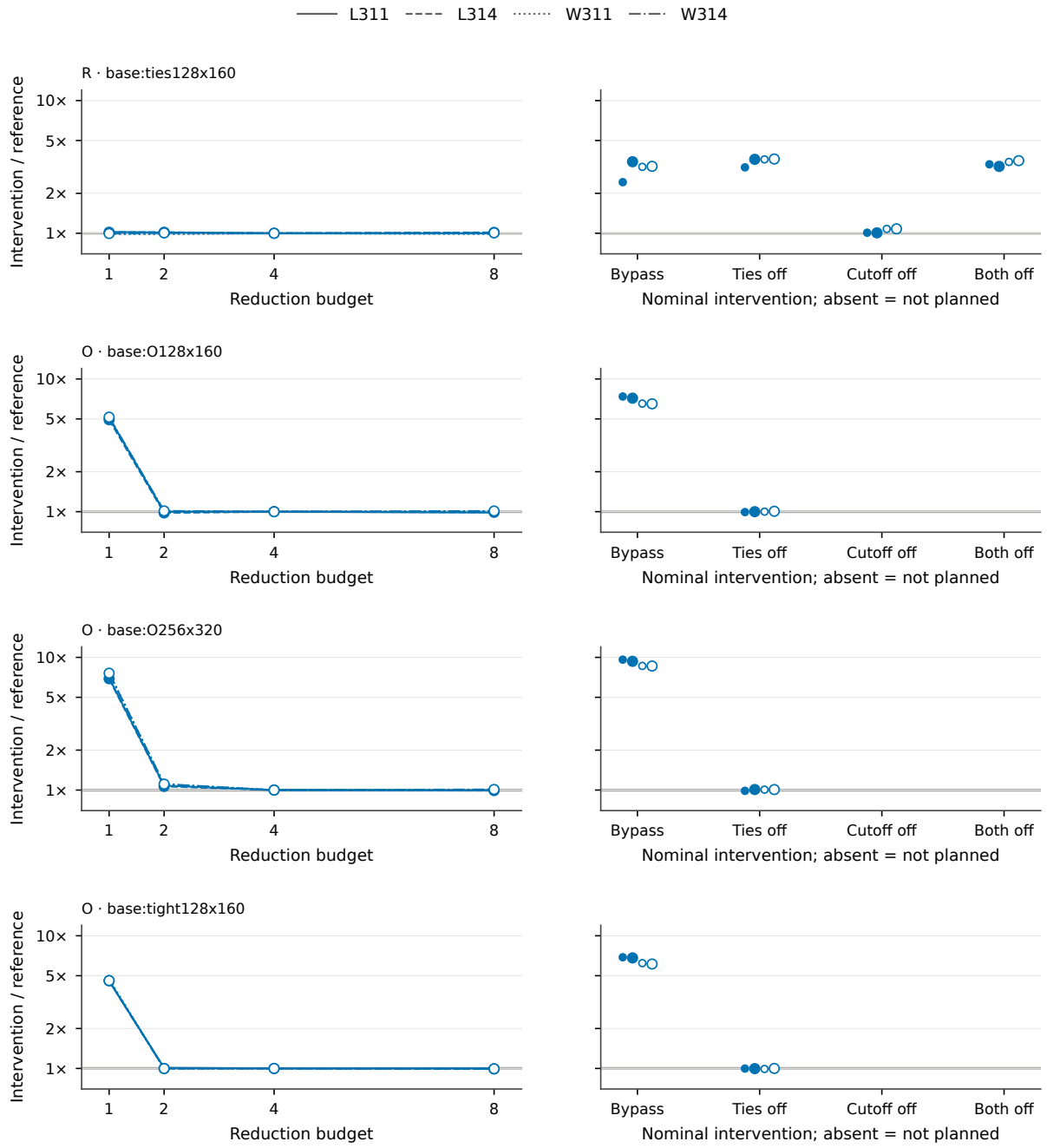
Atlas 21. Rust policy cases, page 1/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



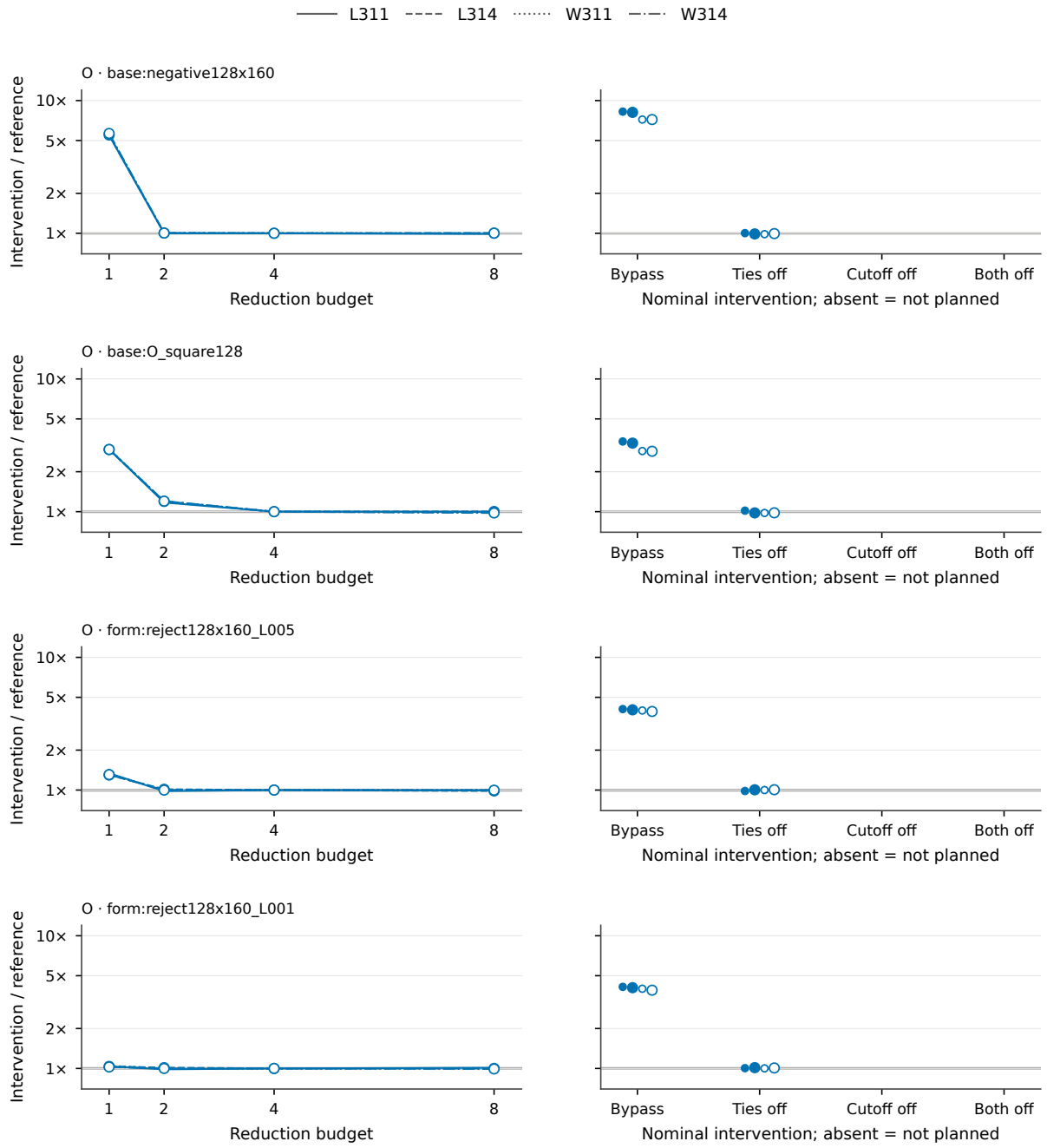
Atlas 22. Rust policy cases, page 2/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



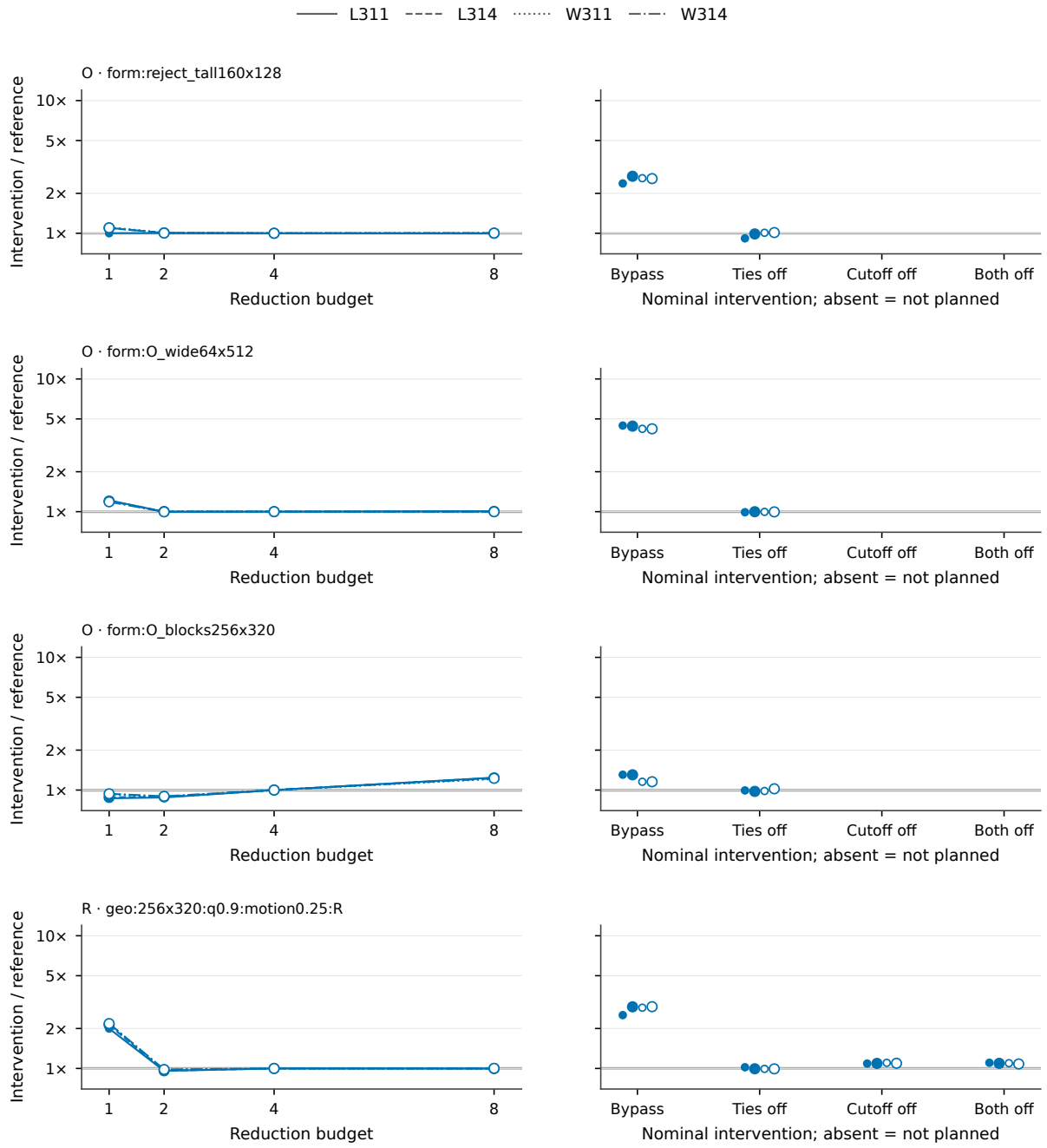
Atlas 23. Rust policy cases, page 3/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



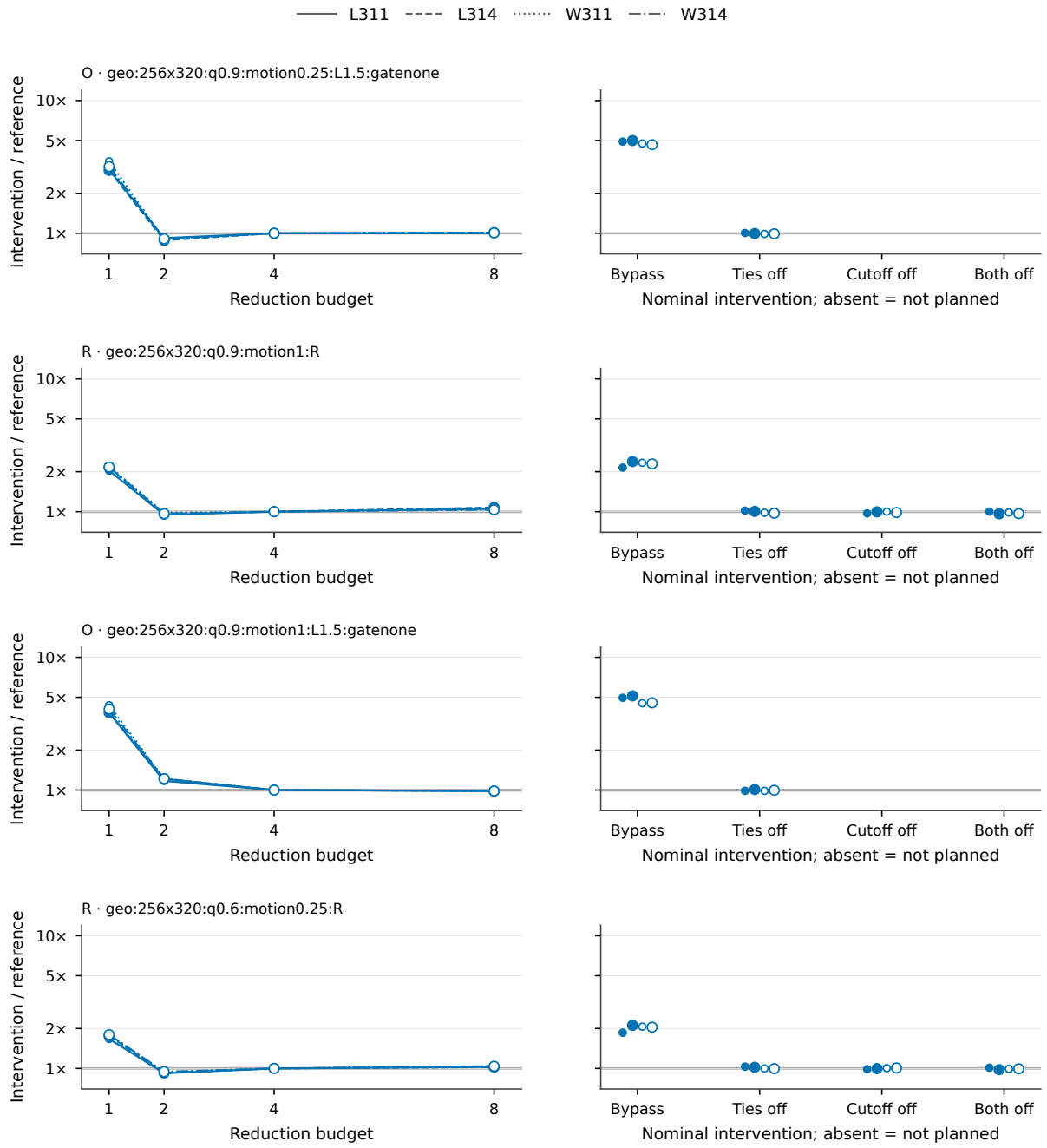
Atlas 24. Rust policy cases, page 4/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



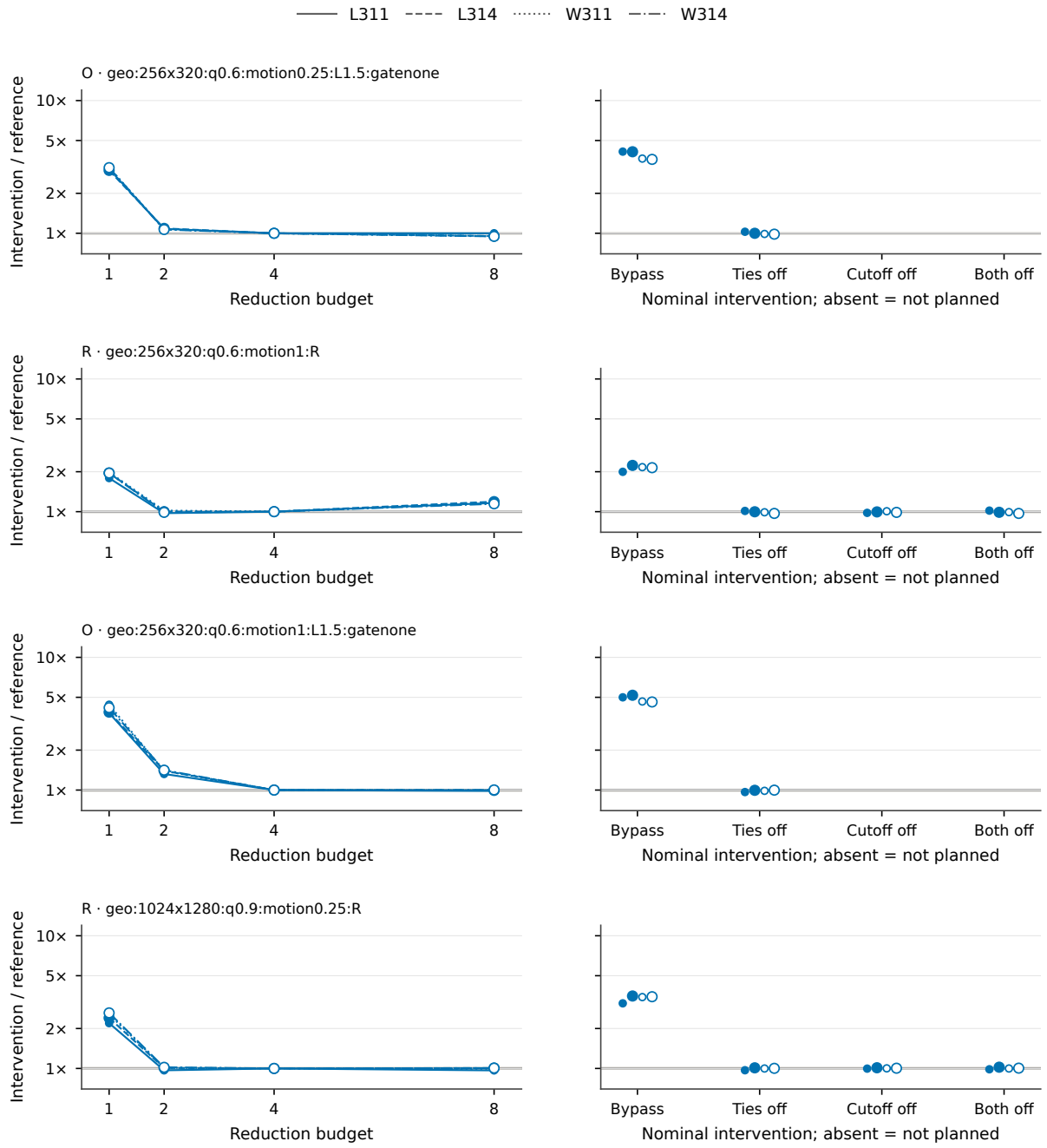
Atlas 25. Rust policy cases, page 5/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



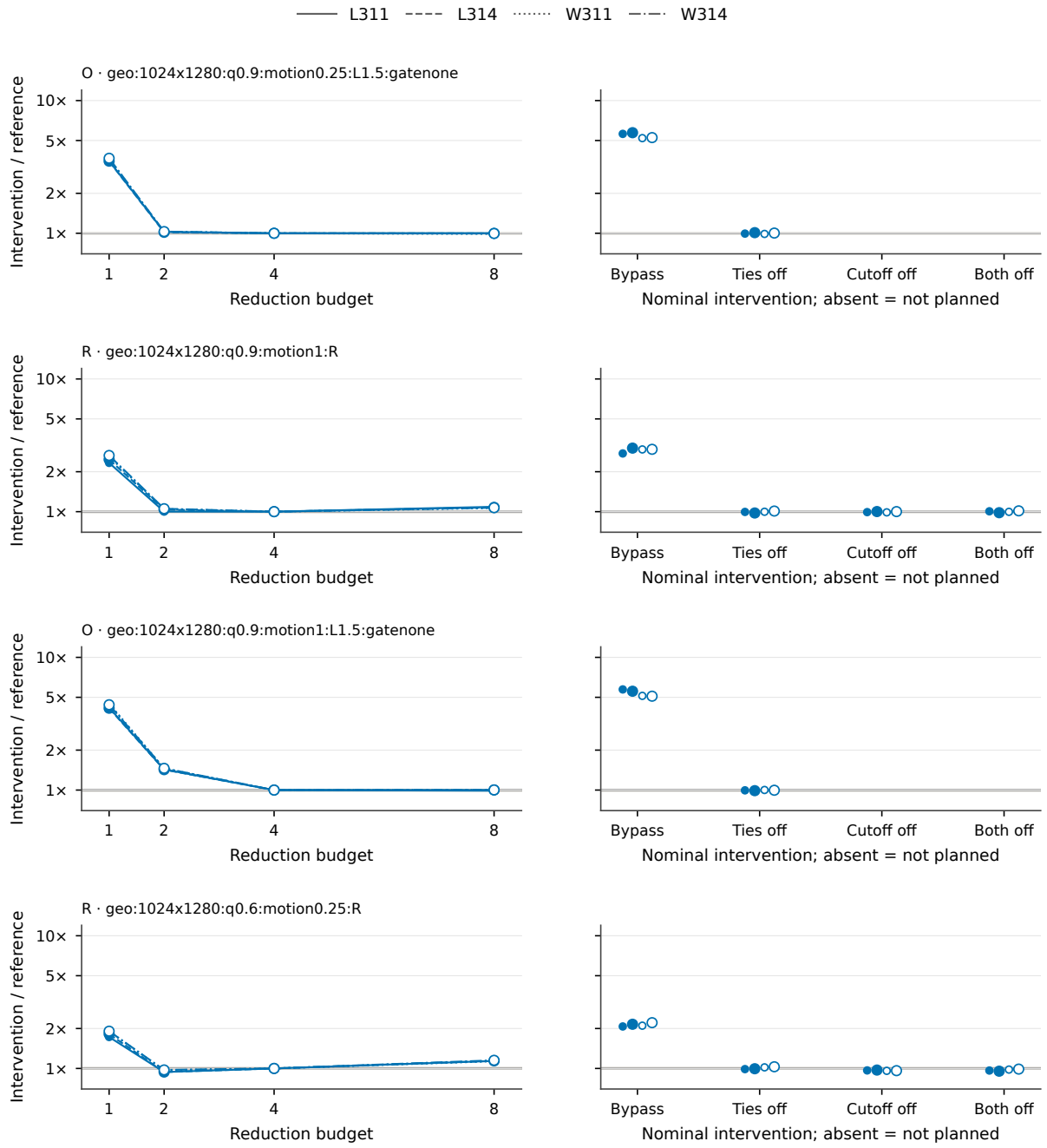
Atlas 26. Rust policy cases, page 6/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



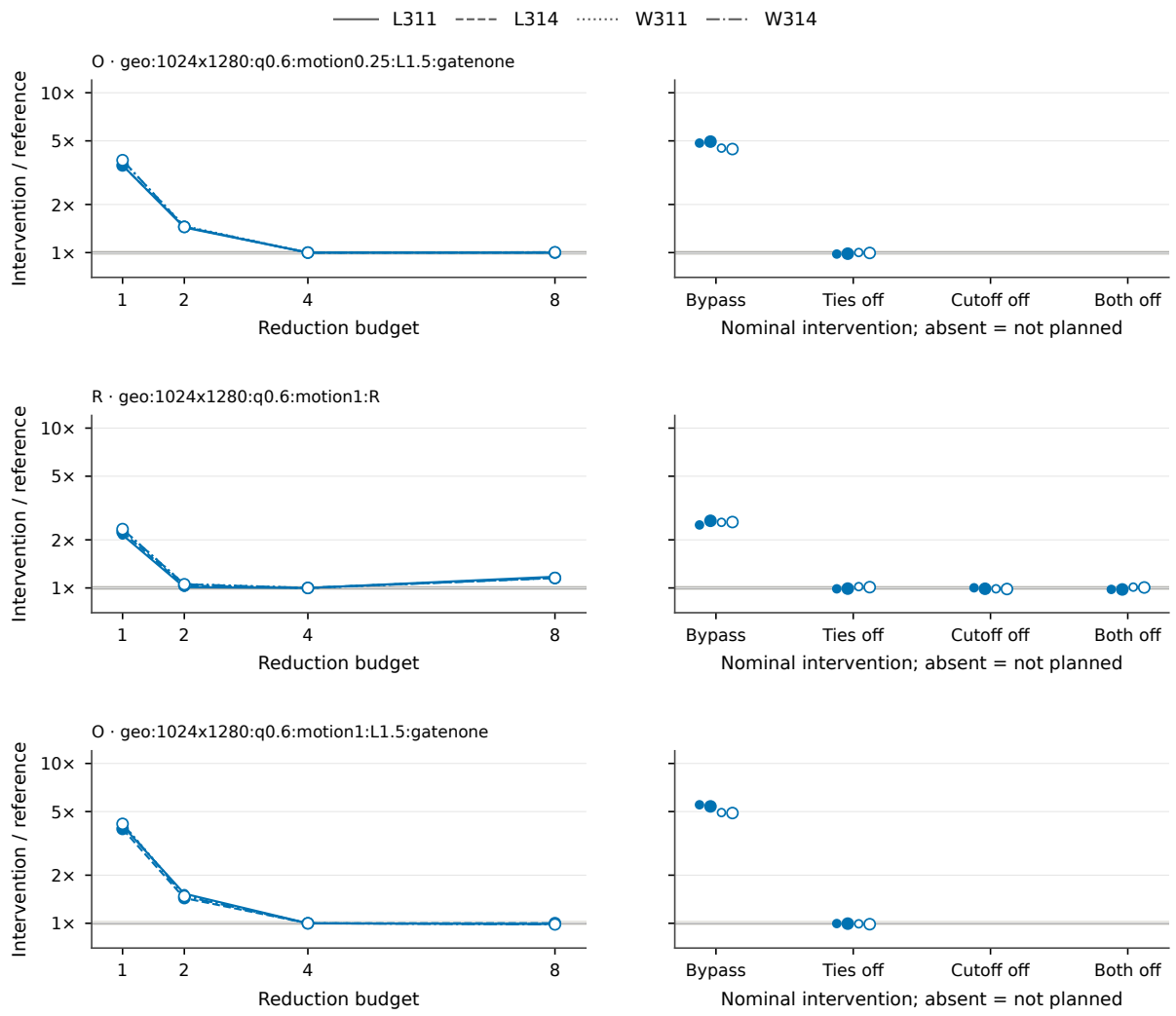
Atlas 27. Rust policy cases, page 7/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



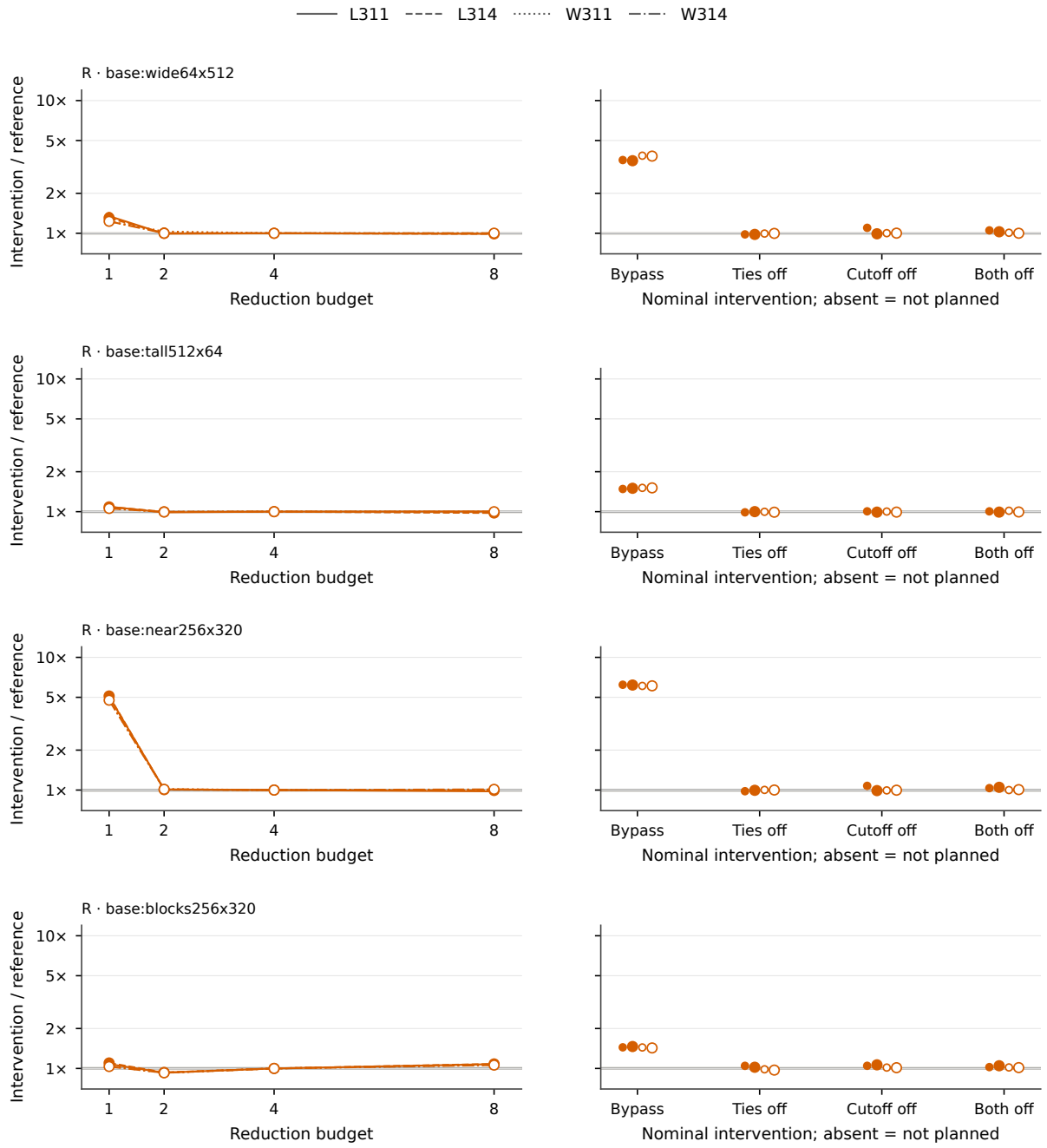
Atlas 28. Rust policy cases, page 8/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



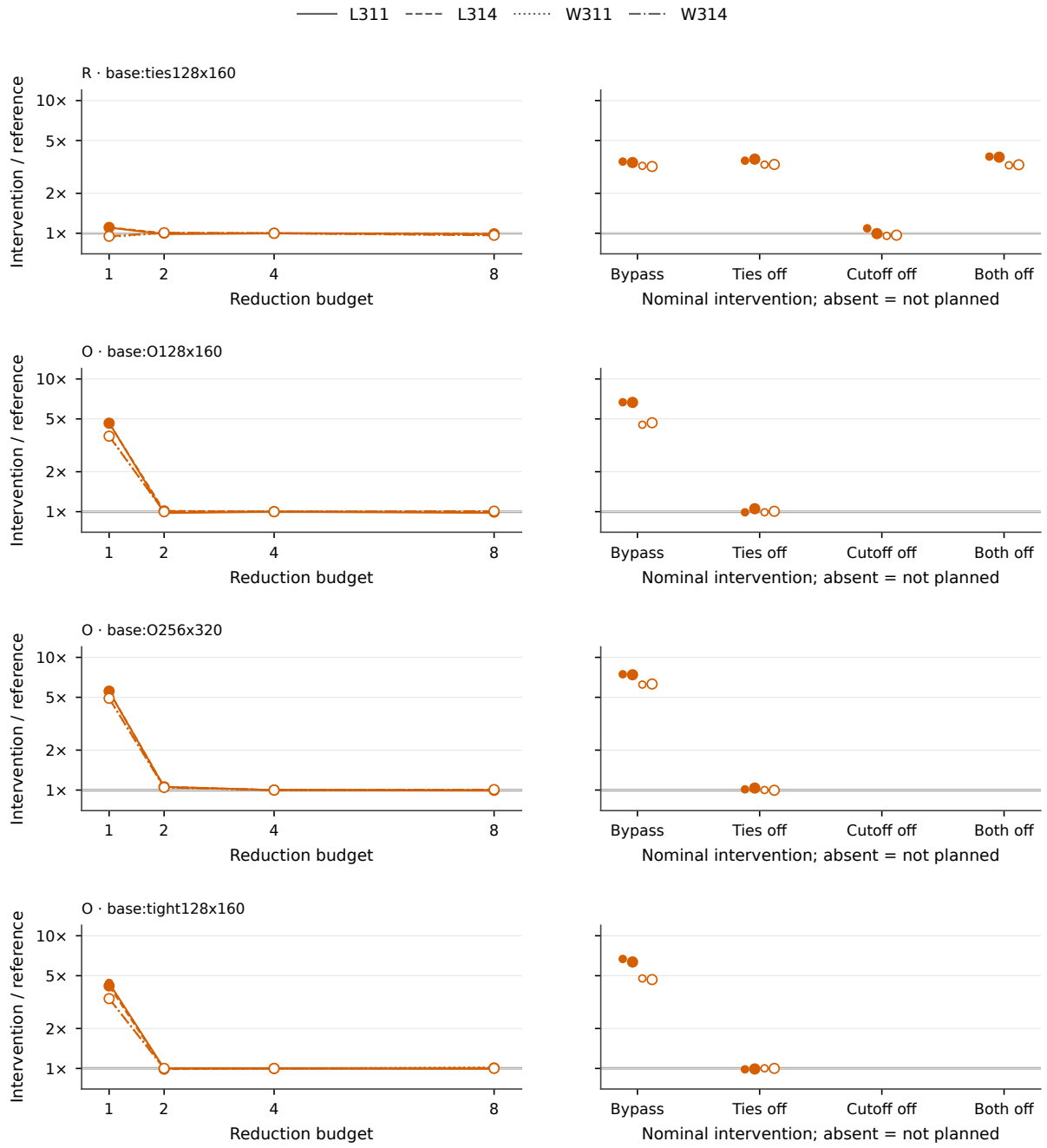
Atlas 29. CPP policy cases, page 1/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



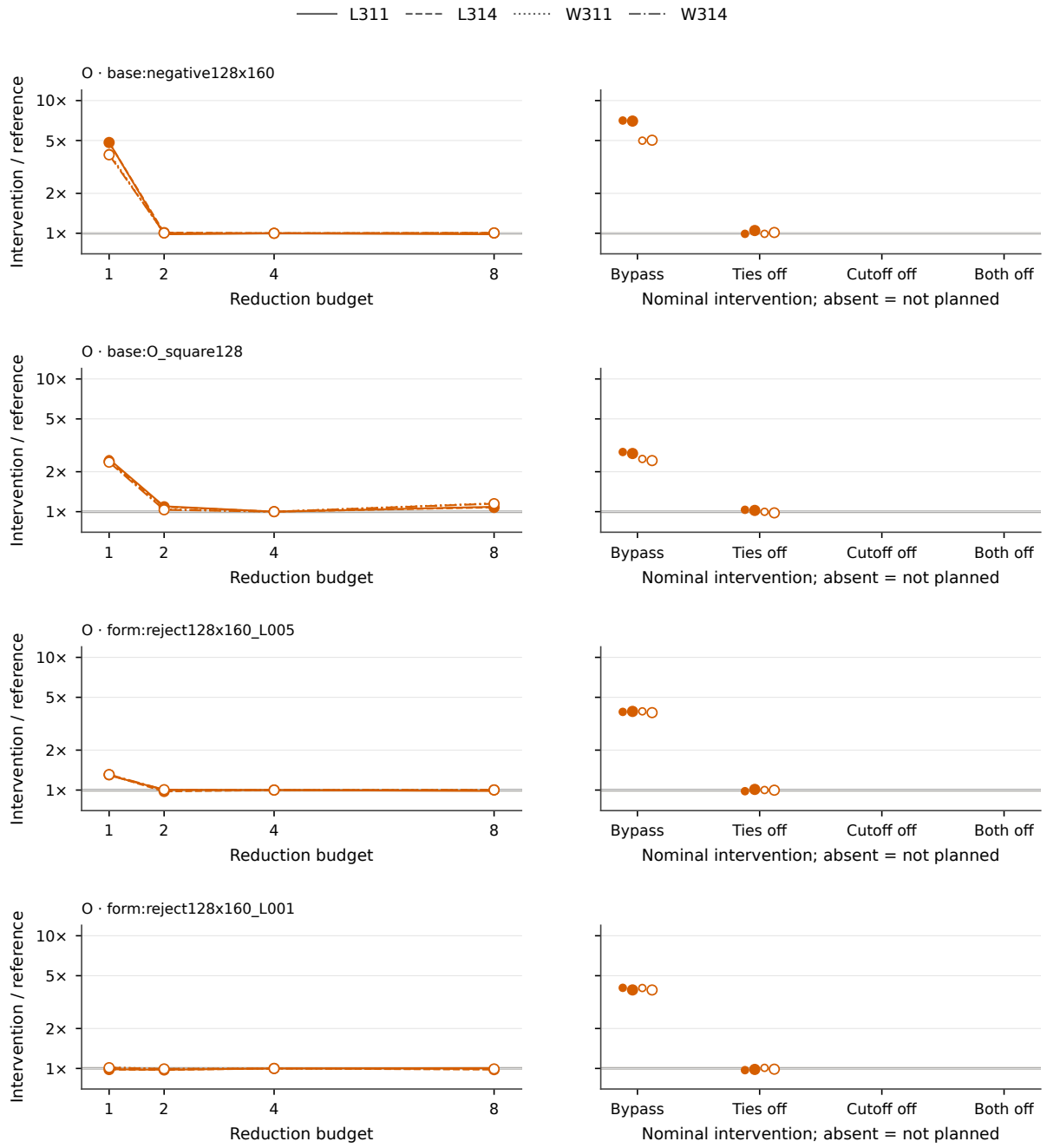
Atlas 30. CPP policy cases, page 2/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



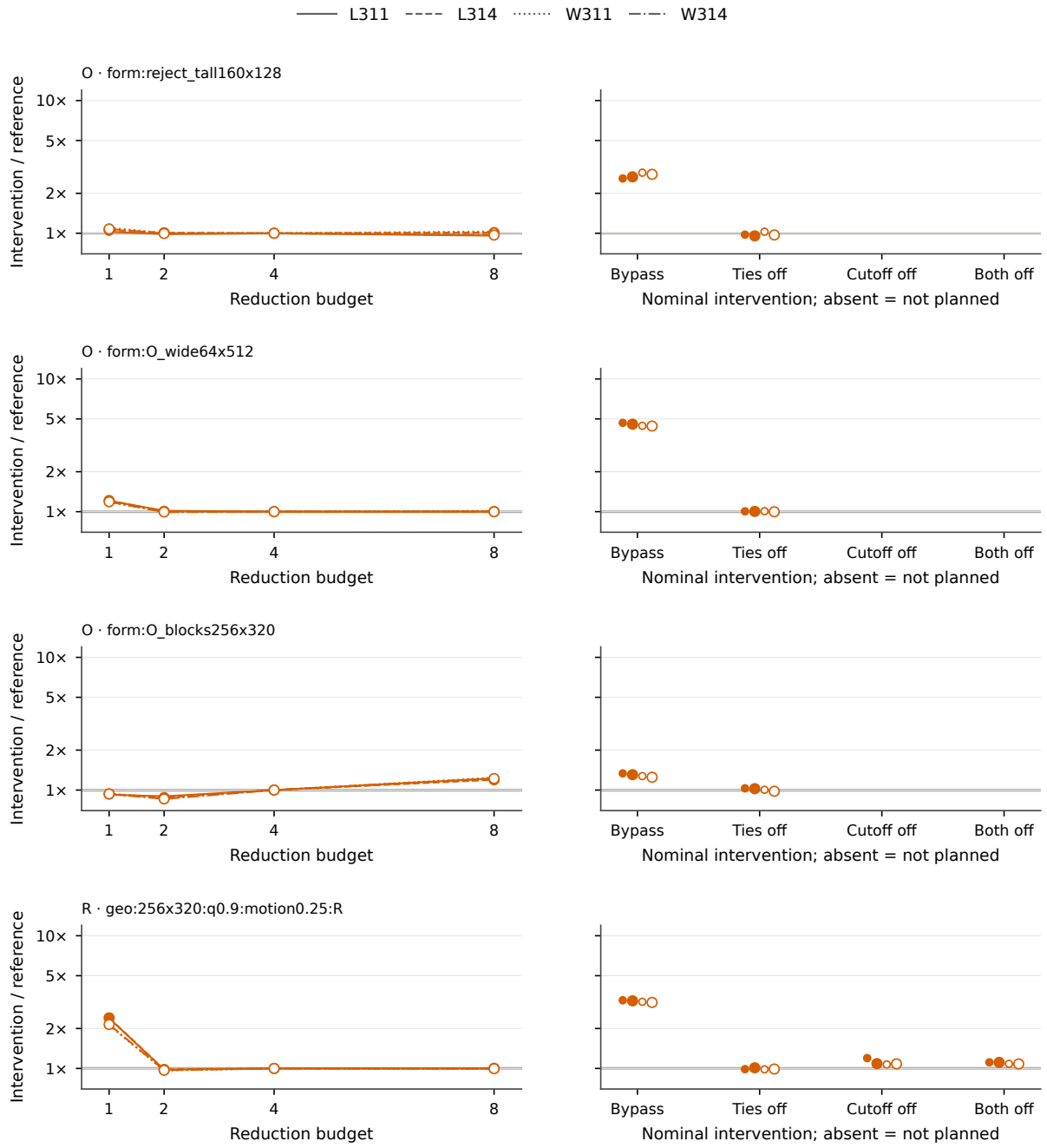
Atlas 31. CPP policy cases, page 3/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



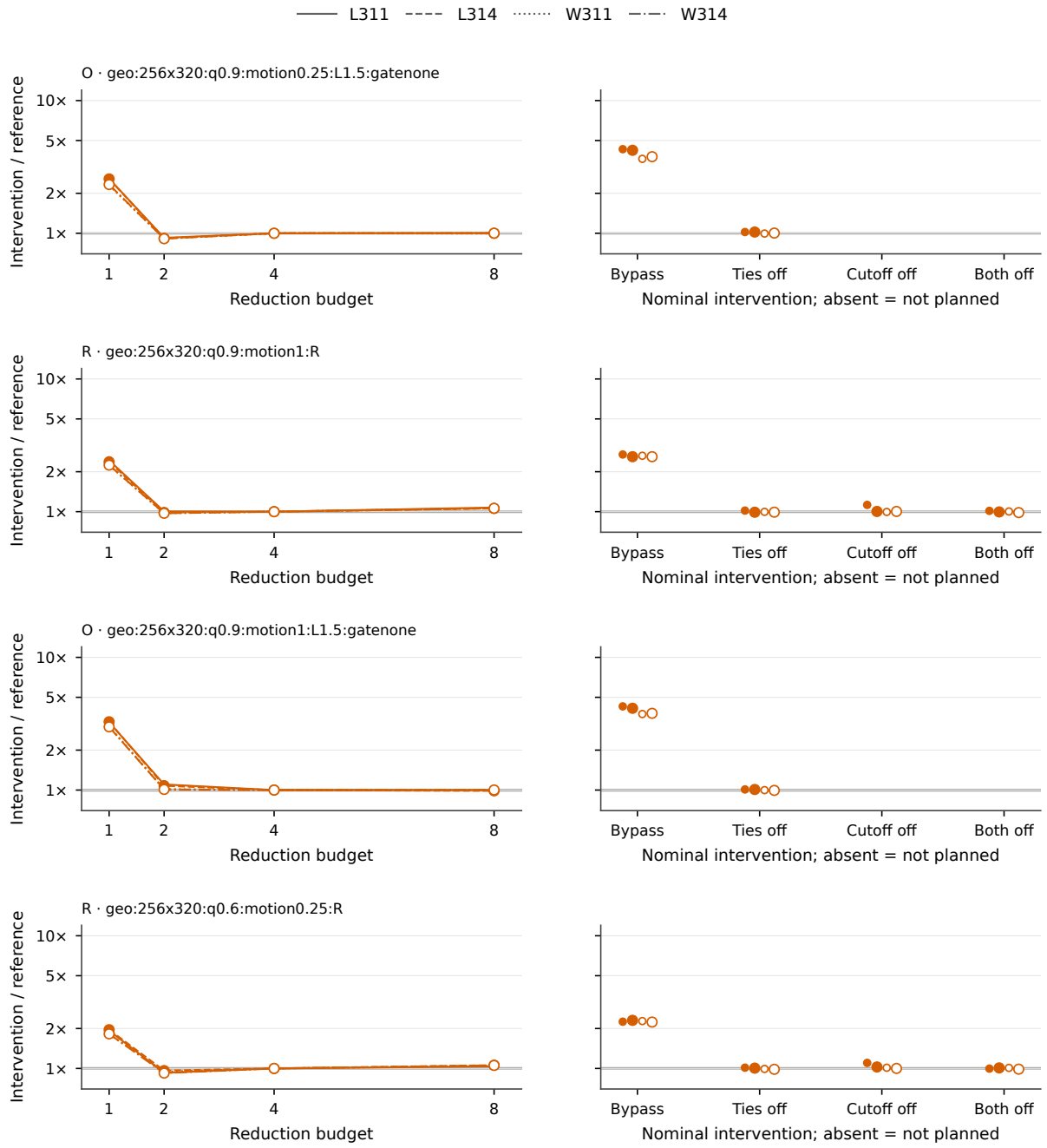
Atlas 32. CPP policy cases, page 4/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



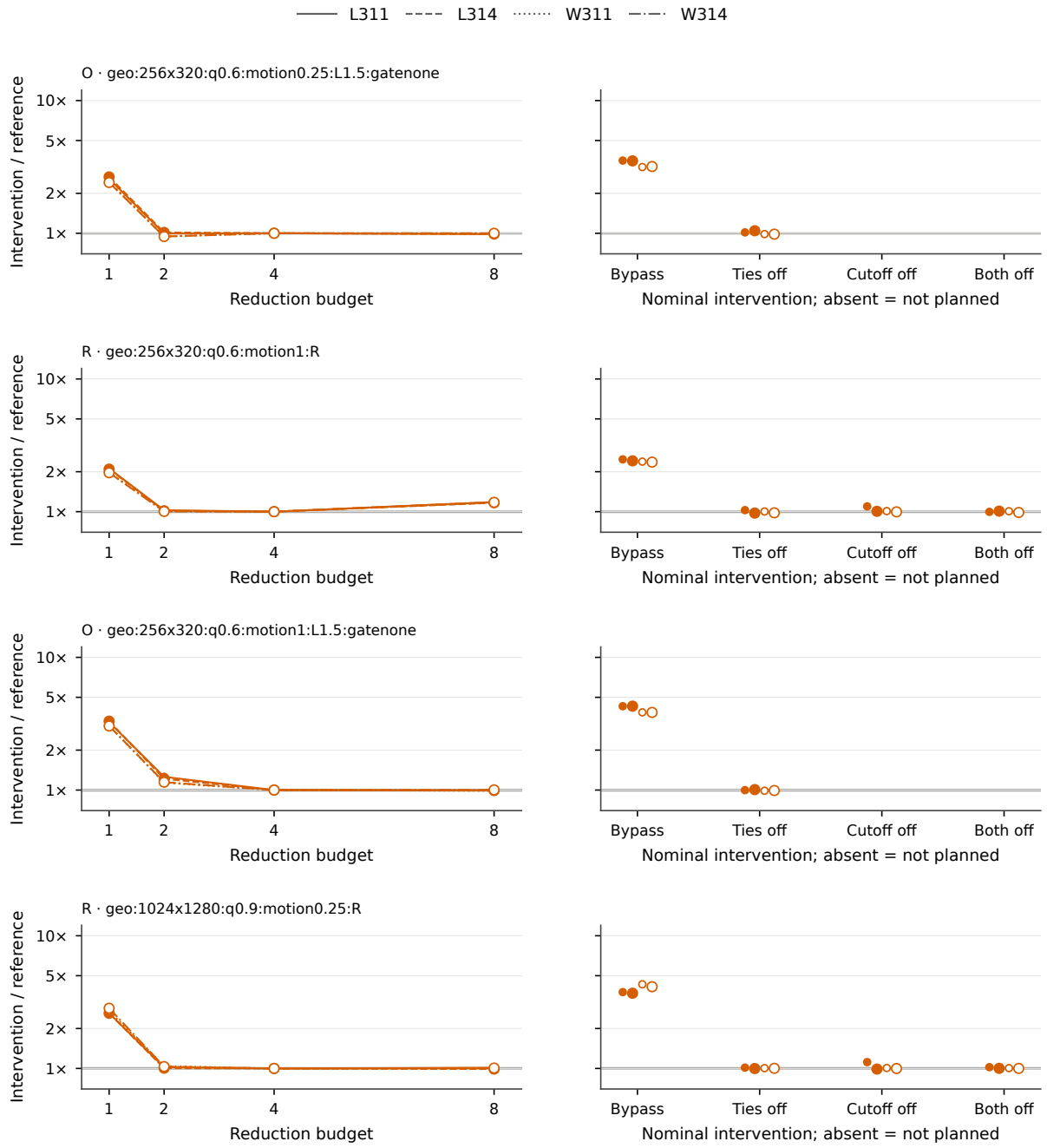
Atlas 33. CPP policy cases, page 5/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



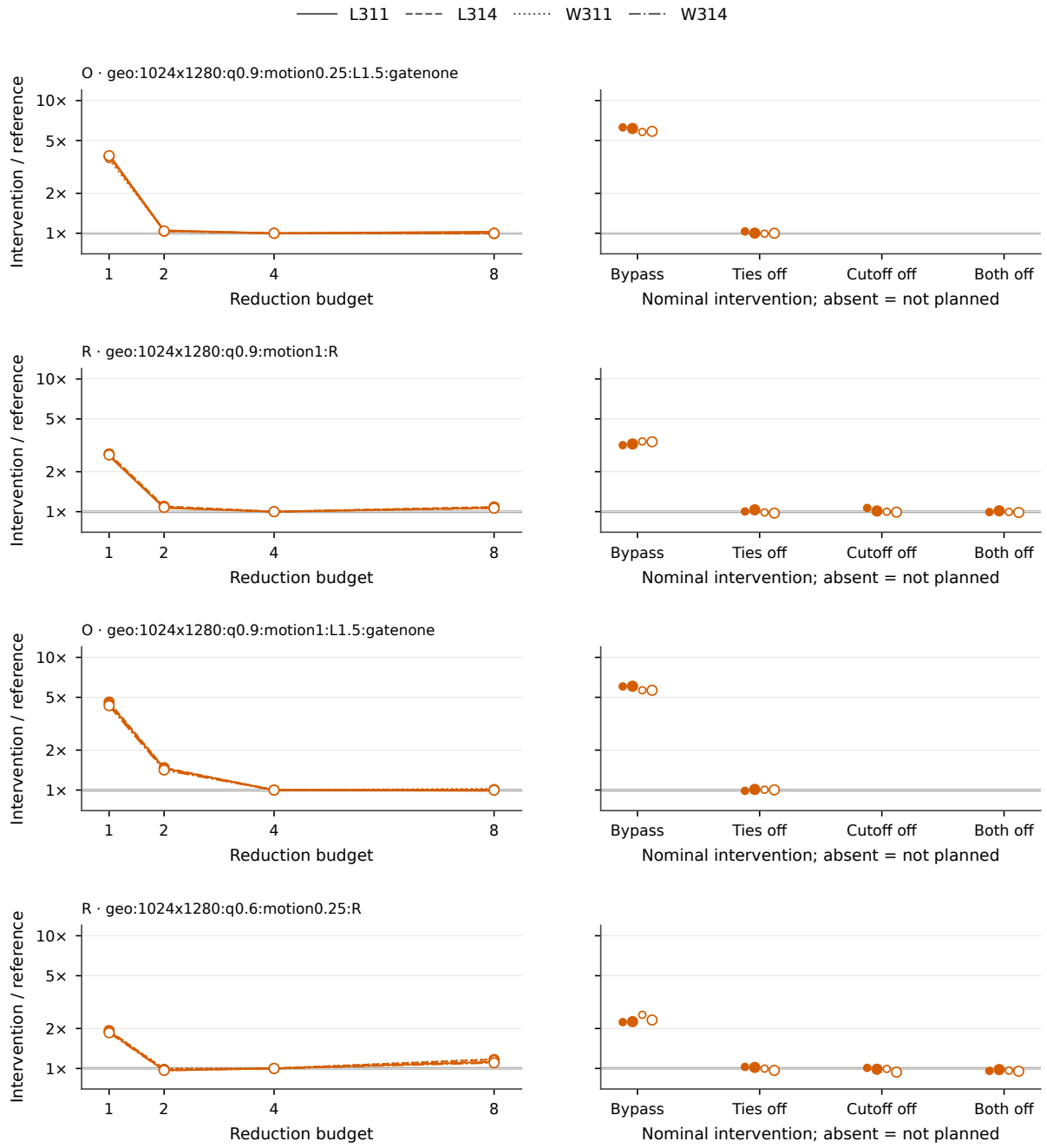
Atlas 34. CPP policy cases, page 6/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



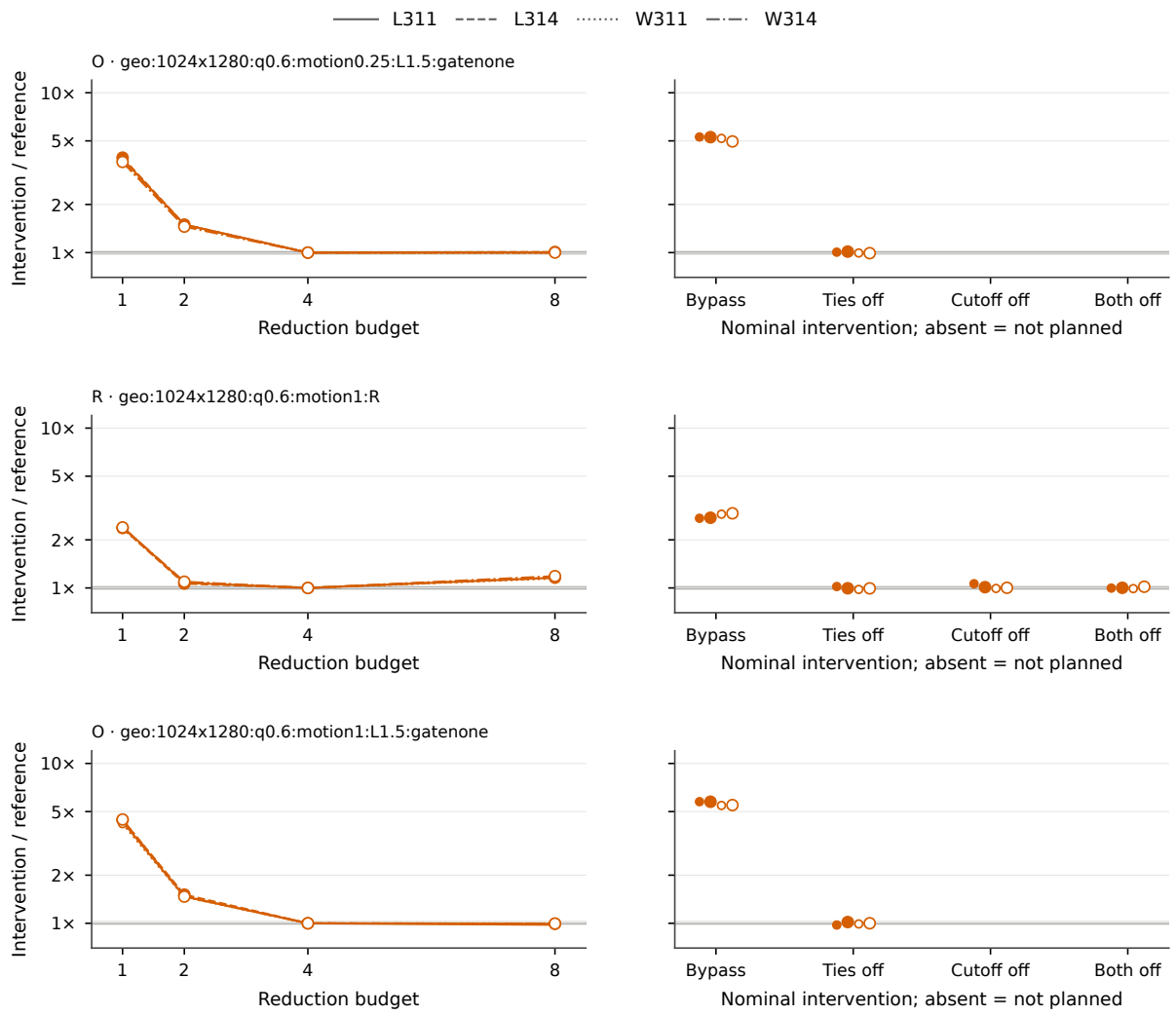
Atlas 35. CPP policy cases, page 7/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



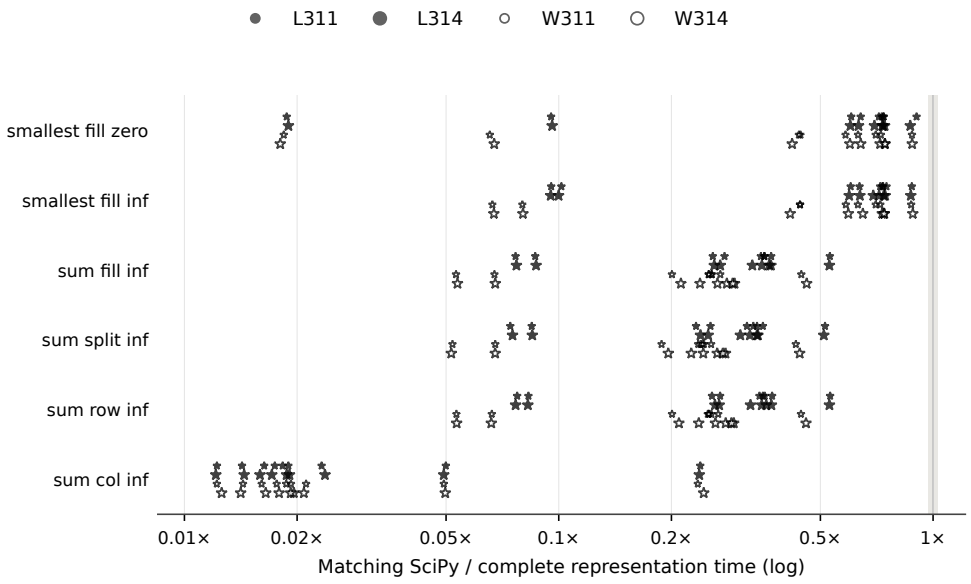
Atlas 36. CPP policy cases, page 8/8

One row per workload; R/O mean required/optional. Left: the ordered reduction-budget sweep. Right: independent nominal interventions, drawn only where planned. Every mark/curve is one environment. Nominal-intervention positions run L311, L314, W311, W314 from left to right. Ratios are intervention/reference, so above one is slower. Exact case identities and all values remain in the indexed S3 policy timing exports and F5_budget_data.csv.



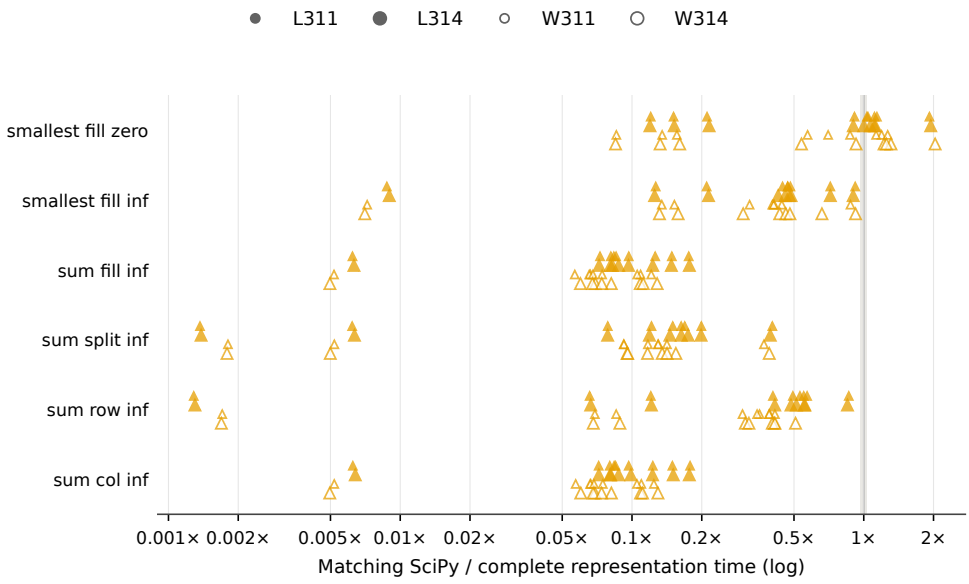
Atlas 37. Shape representations: scipy

All 9 eligible workloads per layout. Each marker is one workload/environment. Within each layout, lanes run L311, L314, W311, W314 from top to bottom. Layouts are categorical and are therefore not connected by lines. Exact case-level values remain in S3_layout_timing.csv.



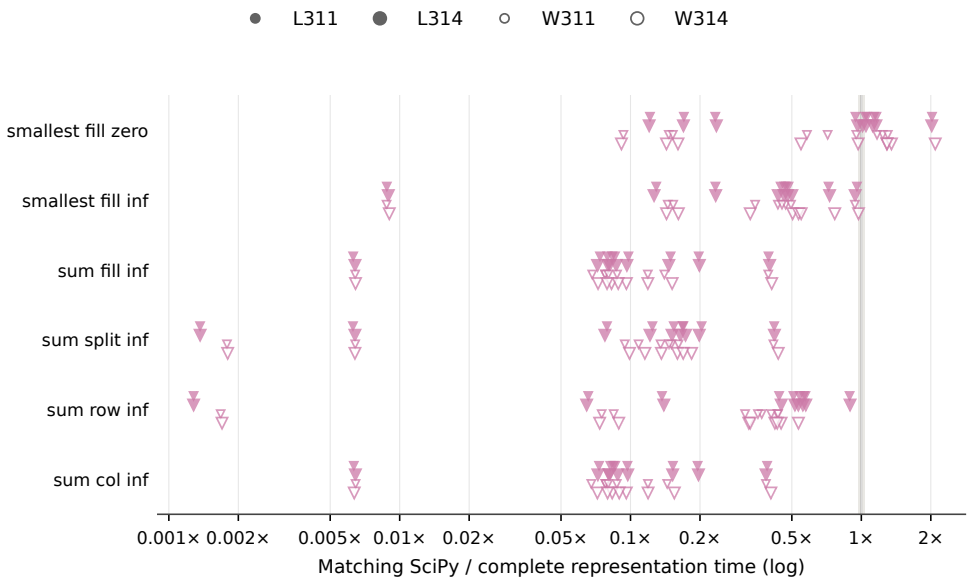
Atlas 38. Shape representations: jvx

All 9 eligible workloads per layout. Each marker is one workload/environment. Within each layout, lanes run L311, L314, W311, W314 from top to bottom. Layouts are categorical and are therefore not connected by lines. Exact case-level values remain in S3_layout_timing.csv.



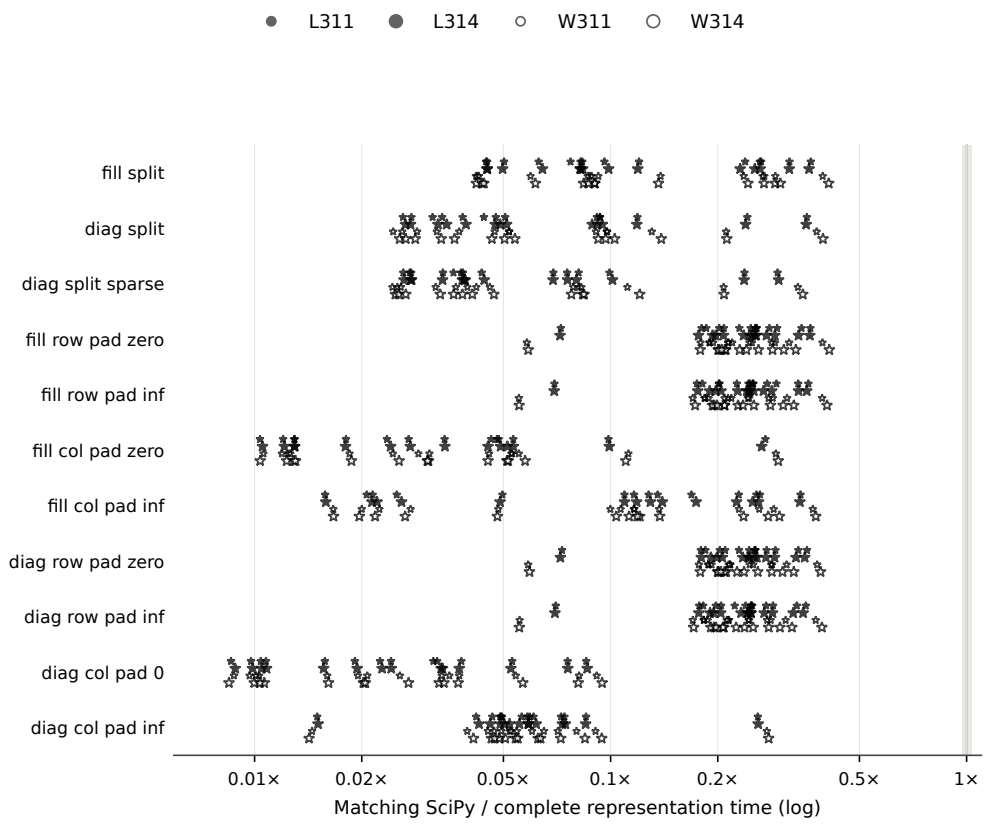
Atlas 39. Shape representations: jvs

All 9 eligible workloads per layout. Each marker is one workload/environment. Within each layout, lanes run L311, L314, W311, W314 from top to bottom. Layouts are categorical and are therefore not connected by lines. Exact case-level values remain in S3_layout_timing.csv.



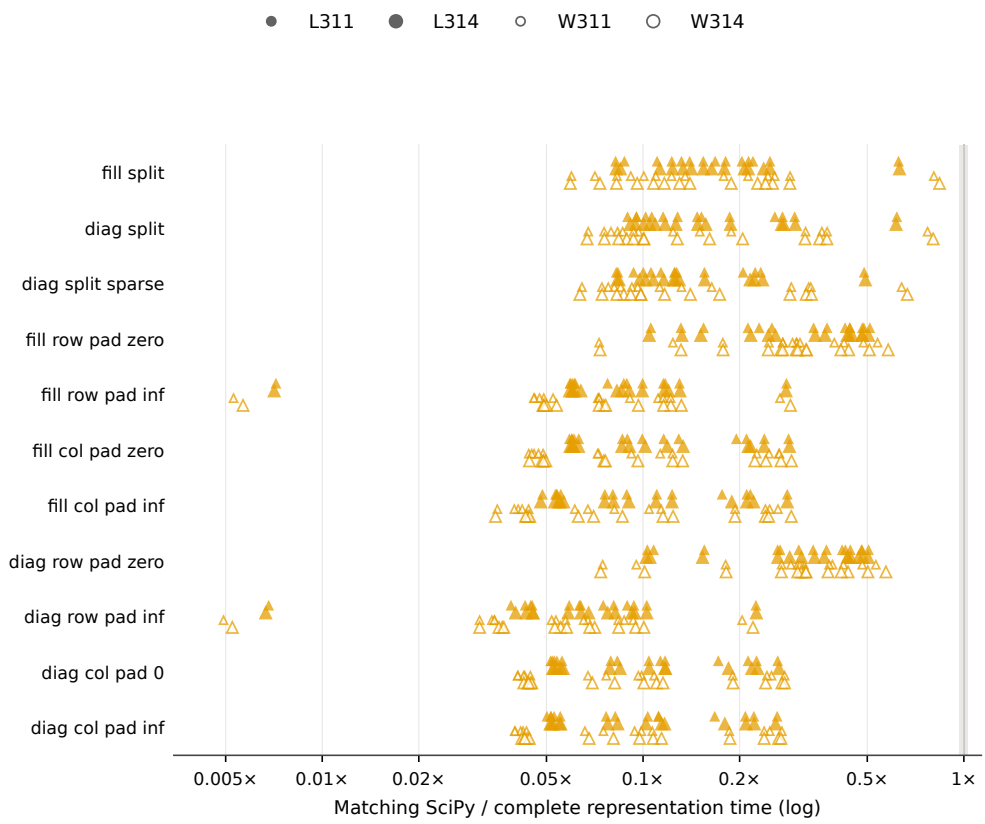
Atlas 40. Cost representations: scipy

All 14 eligible workloads per layout. Each marker is one workload/environment. Within each layout, lanes run L311, L314, W311, W314 from top to bottom. Layouts are categorical and are therefore not connected by lines. Exact case-level values remain in S3_layout_timing.csv.



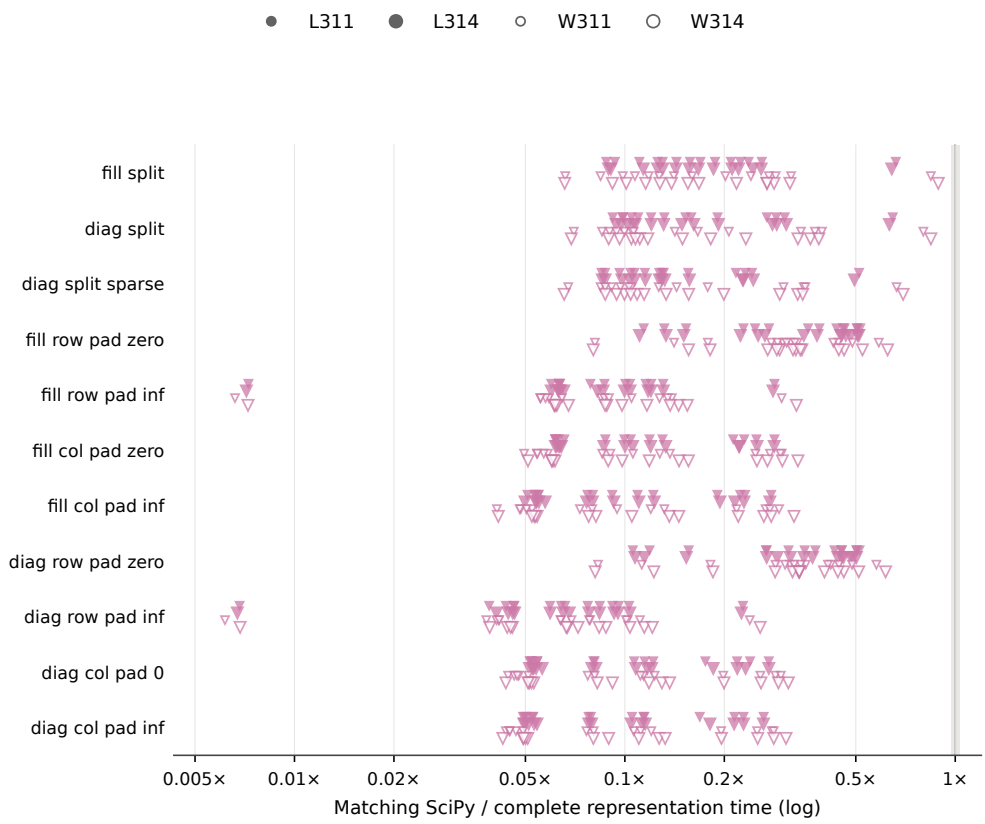
Atlas 41. Cost representations: jvx

All 14 eligible workloads per layout. Each marker is one workload/environment. Within each layout, lanes run L311, L314, W311, W314 from top to bottom. Layouts are categorical and are therefore not connected by lines. Exact case-level values remain in S3_layout_timing.csv.



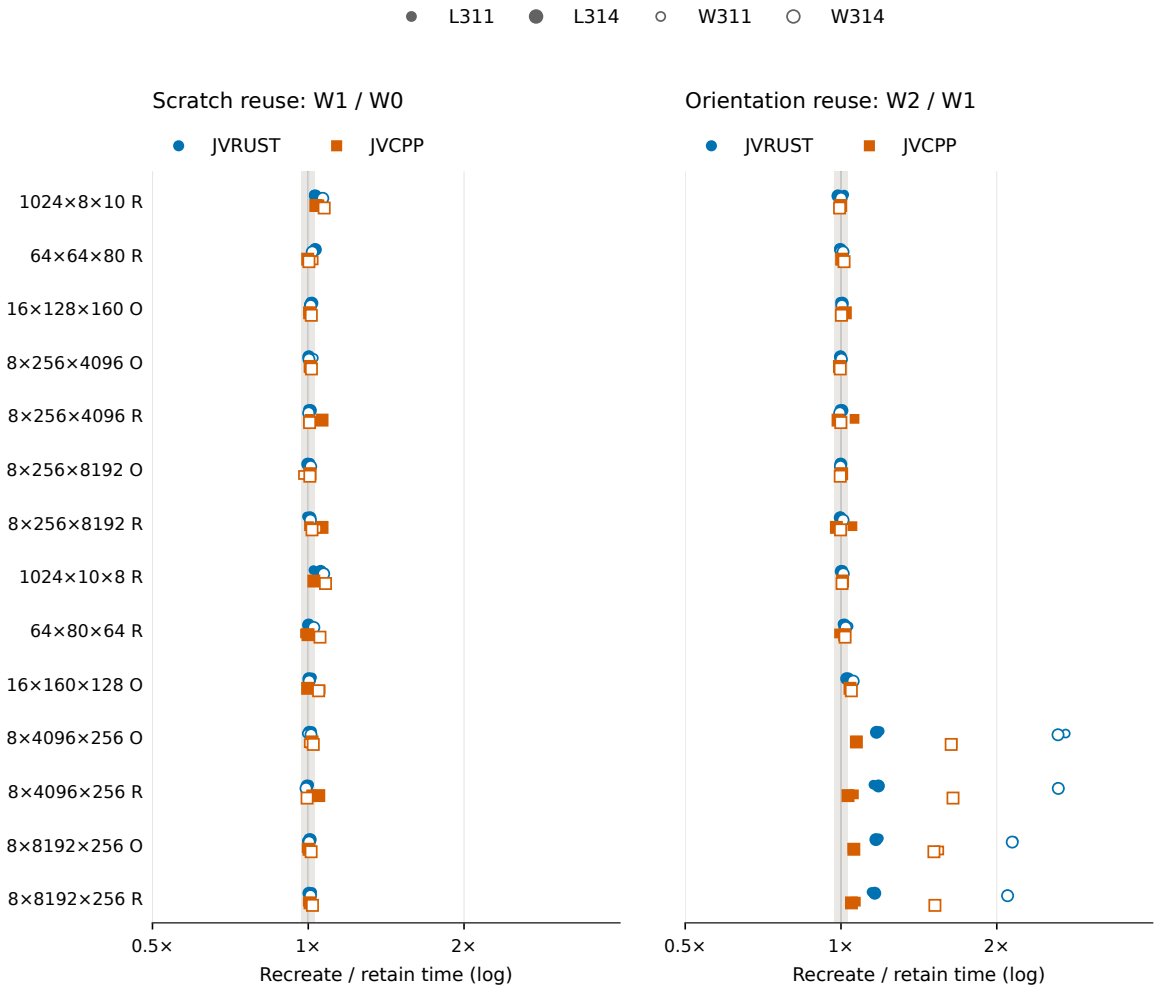
Atlas 42. Cost representations: jvs

All 14 eligible workloads per layout. Each marker is one workload/environment. Within each layout, lanes run L311, L314, W311, W314 from top to bottom. Layouts are categorical and are therefore not connected by lines. Exact case-level values remain in S3_layout_timing.csv.



Atlas 43. Workspace reuse with one worker

Each marker is one workload, method and environment. Filled/open markers identify Linux/Windows; smaller/larger markers identify Python 3.11/3.14. Within each method, lanes run L311, L314, W311, W314 from top to bottom. Recreate/retain ratios above one favour retaining state. The main batch figure shows four-worker reuse.



Application API ratios

Each cell gives float32 assignment replay / tracker.update speed ratios, each relative to SciPy within the same environment. Values are displayed to three decimals; classifications use unrounded values. Absolute times and round estimates remain in S4 exports.

MOT16-04 / GT / Centroid

API	L311	L314	W311	W314
JVRUST	2.505 / 1.084	2.465 / 1.078	2.564 / 1.090	2.574 / 1.085
JVCPP	2.515 / 1.083	2.478 / 1.077	2.380 / 1.084	2.398 / 1.083
JVHYBRID	2.478 / 1.079	2.456 / 1.076	2.614 / 1.086	2.598 / 1.090
JVX row-map	1.355 / 1.017	1.350 / 1.011	1.299 / 1.018	1.258 / 1.013
JVX indices	1.411 / 1.027	1.405 / 1.013	1.367 / 1.022	1.319 / 1.021
JVX pairs	1.574 / 1.041	1.543 / 1.033	1.519 / 1.042	1.491 / 1.029
JVS	1.547 / 1.031	1.572 / 1.018	1.484 / 1.023	1.321 / 1.028

MOT16-04 / RF-DETR / Centroid

API	L311	L314	W311	W314
JVRUST	1.935 / 1.072	1.917 / 1.067	1.992 / 1.078	2.013 / 1.072
JVCPP	1.990 / 1.073	1.978 / 1.069	1.991 / 1.080	2.013 / 1.066
JVHYBRID	1.904 / 1.073	1.886 / 1.060	2.012 / 1.082	2.022 / 1.073
JVX row-map	0.714 / 0.928	0.711 / 0.925	0.752 / 0.942	0.732 / 0.930
JVX indices	0.742 / 0.939	0.734 / 0.932	0.777 / 0.954	0.763 / 0.940
JVX pairs	0.816 / 0.966	0.794 / 0.958	0.870 / 0.977	0.848 / 0.963
JVS	0.732 / 0.934	0.749 / 0.937	0.783 / 0.944	0.775 / 0.926

MOT20-03 / GT / Centroid

API	L311	L314	W311	W314
JVRUST	2.778 / 1.095	2.778 / 1.091	2.922 / 1.111	2.878 / 1.109
JVCPP	2.809 / 1.096	2.807 / 1.095	2.654 / 1.107	2.699 / 1.108
JVHYBRID	2.803 / 1.095	2.800 / 1.094	2.922 / 1.113	2.917 / 1.114
JVX row-map	1.709 / 1.050	1.709 / 1.046	1.653 / 1.060	1.662 / 1.047
JVX indices	1.736 / 1.054	1.737 / 1.048	1.731 / 1.059	1.700 / 1.055
JVX pairs	1.810 / 1.061	1.800 / 1.058	1.819 / 1.068	1.780 / 1.067
JVS	1.929 / 1.061	1.957 / 1.056	1.887 / 1.069	1.859 / 1.062

MOT20-03 / RF-DETR / Centroid

API	L311	L314	W311	W314
JVRUST	2.136 / 1.078	2.109 / 1.077	2.229 / 1.087	2.168 / 1.086
JVCPP	2.213 / 1.077	2.195 / 1.079	2.232 / 1.092	2.206 / 1.087
JVHYBRID	2.098 / 1.071	2.060 / 1.075	2.263 / 1.091	2.229 / 1.078
JVX row-map	0.663 / 0.923	0.658 / 0.918	0.783 / 0.953	0.760 / 0.946
JVX indices	0.677 / 0.926	0.671 / 0.924	0.804 / 0.962	0.782 / 0.951
JVX pairs	0.714 / 0.941	0.700 / 0.941	0.861 / 0.976	0.841 / 0.969
JVS	0.677 / 0.925	0.682 / 0.927	0.828 / 0.954	0.813 / 0.955

MOT16-04 / GT / Sort

API	L311	L314	W311	W314
JVRUST	2.427 / 1.005	2.389 / 1.005	2.524 / 1.004	2.492 / 1.000
JVCPP	2.504 / 1.006	2.471 / 1.009	2.406 / 1.006	2.397 / 1.004
JVHYBRID	2.400 / 1.005	2.384 / 1.006	2.561 / 0.997	2.519 / 1.005
JVX row-map	1.354 / 0.999	1.350 / 0.999	1.312 / 0.996	1.248 / 0.993
JVX indices	1.411 / 1.000	1.407 / 0.999	1.364 / 1.001	1.309 / 0.999
JVX pairs	1.567 / 1.000	1.542 / 1.003	1.513 / 1.000	1.457 / 0.996
JVS	1.513 / 1.000	1.549 / 1.002	1.470 / 1.006	1.412 / 0.999

MOT16-04 / RF-DETR / Sort

API	L311	L314	W311	W314
JVRUST	2.290 / 1.014	2.261 / 1.008	2.301 / 1.008	2.314 / 1.004
JVCPP	2.329 / 1.015	2.316 / 1.008	2.280 / 0.998	2.287 / 1.005
JVHYBRID	2.239 / 1.015	2.218 / 1.007	2.339 / 1.000	2.328 / 1.010
JVX row-map	0.847 / 0.999	0.838 / 0.989	0.834 / 0.992	0.804 / 0.988
JVX indices	0.880 / 0.998	0.873 / 0.989	0.860 / 0.993	0.837 / 0.991
JVX pairs	0.970 / 1.002	0.952 / 0.992	0.965 / 0.995	0.947 / 0.990
JVS	0.866 / 0.999	0.891 / 0.992	0.865 / 0.994	0.845 / 0.992

MOT20-03 / GT / Sort

API	L311	L314	W311	W314
JVRUST	2.930 / 1.010	2.923 / 1.011	3.143 / 1.012	3.073 / 1.005
JVCPP	2.990 / 1.012	2.976 / 1.011	2.919 / 1.009	2.908 / 1.009
JVHYBRID	2.941 / 1.010	2.935 / 1.011	3.172 / 1.011	3.069 / 1.008
JVX row-map	1.908 / 1.002	1.904 / 1.005	1.802 / 1.003	1.798 / 1.002
JVX indices	1.946 / 1.006	1.944 / 1.006	1.911 / 1.007	1.834 / 1.001
JVX pairs	2.039 / 1.007	2.021 / 1.008	2.007 / 1.008	1.921 / 1.003
JVS	2.147 / 1.007	2.175 / 1.008	2.094 / 1.006	2.017 / 1.002

MOT20-03 / RF-DETR / Sort

API	L311	L314	W311	W314
JVRUST	2.659 / 1.004	2.625 / 1.004	2.691 / 0.995	2.693 / 1.006
JVCPP	2.702 / 1.005	2.676 / 1.005	2.669 / 0.995	2.649 / 1.006
JVHYBRID	2.585 / 1.005	2.550 / 1.004	2.736 / 1.006	2.717 / 1.005
JVX row-map	1.097 / 0.997	1.084 / 0.996	1.052 / 0.992	1.028 / 0.996
JVX indices	1.129 / 0.997	1.118 / 0.996	1.099 / 0.987	1.069 / 1.000
JVX pairs	1.228 / 0.997	1.197 / 0.997	1.198 / 0.991	1.167 / 0.999
JVS	1.136 / 0.997	1.152 / 0.997	1.127 / 0.988	1.100 / 0.999